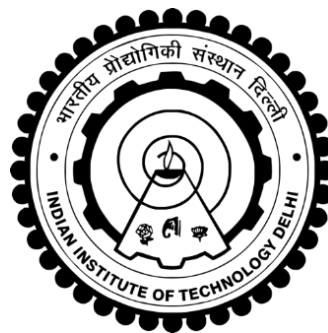


MCP 101

Product Realization by Manufacturing Lab Manual

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Department of Mechanical Engineering
INDIAN INSTITUTE OF TECHNOLOGY DELHI
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Chapter 1

Machine Shop

In a machine shop, metals are cut to shape on different machine tools. The work is performed due to relative motion between the job and tool. A machine tool generally consists of a few parts as follows: Base, Power source with arrangement of gear box or other power transferring mechanism, Work piece holding component like chuck, table, vice, etc., Tool holding device like tool post, arbor etc.

The following are the machine tools which are used in machine shops for different types of operations and shapes.

Lathe machine: A lathe is used to cut and shape the metal by revolving the work against a cutting tool. Primarily, this machine is used for turning operations. The work is clamped either in a chuck, fitted on to the lathe spindle or in-between the centers. The cutting tool is fixed in a tool post, mounted on a movable carriage that is positioned on the lathe bed. The cutting tool can be fed on to the work, either lengthwise or crosswise. While turning, the chuck rotates in a counterclockwise direction, when viewed from the tail stock end.

Shaper machine: Shapers are intended to produce horizontal, vertical, or inclined flat surfaces by means of a reciprocating single point cutting tool. Tool is given reciprocating motion with the help of mechanism. In the shaper the tool is held in the tool post of reciprocating ram and the work piece is held in a vice. Cutting takes place in the forward stroke and no material is removed in the return stroke. The work piece is given indexed feed perpendicular to the direction of ram at the end of return stroke. To reduce time in the return stroke, the Whitworth quick return mechanism is used, which results in the mean speed of the return stroke being higher than that of the forward stroke.

Planer machine: Planer is used to produce large flat surfaces which are almost impracticable to be machined on a shaper. In planer the work piece reciprocates past the stationary single point cutting tool as compared to shaper in which the cutting tool reciprocates past the stationary work.

Slotter machine: The vertical position of a shaper machine is known as slotting machine which is used to produce splines, slots etc. The slotting machine is a reciprocating machine tool in which the ram holding the tool reciprocates in a vertical axis and the cutting action of the tool is only during the downward stroke.

Drilling machine: Drilling is an operation of producing circular holes in a work piece by using a rotating tool known as drill. In drilling operation, a hole is produced by feeding the rotating drill in a direction parallel to its axis into a work piece fixed to the table. During cutting, the metal comes in contact with the two cutting edges of the tool and removes the metal stock.

Drive Mechanism of Drilling Machine: When the power is given to the motor, the spindle rotates and thereby the stepped pulley attached to it also rotates. On the other end, one more stepped pulley is attached and that is inverted to increase or decrease the speed of the rotational motion.

Now, a V-belt is placed in between the stepped pulleys to drive the power transmission. Here a V-belt is used instead of a flat belt, to increase the power efficiency.

Now the drill bit also rotates which was placed in the chuck and which was in connection with the spindle. As the Pulleys rotates, the spindle also rotates which can rotate the drill bit.

Now, by the rotation of hand-wheel, the spindle moves up and down in the vertical direction to give the necessary amount of feed to the work and this drill bit is used to make the holes in the component placed in the machine vice.

The conversion of the rotary motion to the linear motion of the spindle is achieved by a rack and pinion mechanism. The rack consists of fine grooves, a pinion (small gear), chuck and the drill bit which is held in the chuck. The chuck is connected to the rack and the rotation of the hand-wheel places the pinion in the rack grooves. This results in converting the rotary motion of the pinion to the linear motion of the rack, which drives the chuck holding the drill bit. Thus, by controlling the hand-wheel, the necessary amount of feed can be provided to the drill bit.

Grinding Machine: Grinding is the metal cutting process in which the material is removed from the work piece by the abrasive action of a rotating cutting tool known as grinding wheel. In grinding wheel abrasive grains are held together by a binding material called bond. Abrasive grains of the grinding wheel act as cutting tool and remove the material in the form of small chips. Grinding is used in finishing operations because it imparts high surface finish by removing a very small amount of material. A low material removal rate is maintained as grinding is a finishing operation and high material removal rates can damage the surface finish of the job. Common abrasives used for grinding are cubic boron nitride (CBN), diamond, aluminum oxide and silicon carbide.

Surface grinding machine: Surface grinders are used to produce flat surfaces. According to the type of table these can be classified as planer type or rotary type. In planer type grinder is rectangular and may have horizontal or vertical spindles.

Power saw: Power saw is power operated saw which is used to shear the metal in large quantity in a short time. They are used to cut large sizes (sections) of metals such as steel. Cutting diameters of more than 10 to 15mm is very hard work with a normal handheld hacksaw. Therefore, power hacksaws have been developed to carry out difficult and time-consuming work. The heavy 'arm' moves backwards and forwards, cutting on the backwards stroke.

Milling machine: The milling machine is used to produce, mainly prismatic components, flat, curved, parallel, stepped, square and inclined faces as well as slots, grooves, threads, and tooth systems. The milling cutter performs a rotary movement (primary motion) and the workpiece a linear movement (secondary motion). Gears are commonly cut in milling machines.

Coolants: Coolants are important during machining operations, especially when a high material removal rate is needed. Coolants cool the workpiece and the tool, reduce friction between the two, reduce wear and tear of the tool, remove the chips formed during machining and thus saving the workpiece from being damaged. Thus, by facilitating easy material removal and dissipating heat, the coolant increases tool life.

Detail of Lathe Machine: Lathe is one of the most versatile and widely used machine tools all over the world. It is commonly known as the mother of all other machine tools. The main function of a lathe is to remove metal from a job to give it the required shape and size. The job is

securely and rigidly held in the chuck or in between centers on the lathe machine and then turn it against a singlepoint cutting tool which will remove metal from the job in the form of chips. Fig. shows the working principle of lathe. An engine lathe is the most basic and simplest form of the lathe. It derives its name from the early lathes, which obtained their power from engines. Besides the simple turning operation as described above, lathe can be used to carry out other operations also, such as drilling, reaming, boring, taper turning, knurling, screw thread cutting, grinding etc.

The term “engine” is associated with this lathe due to the fact that in the very early days of its development it was driven by a steam engine. This lathe is an important member of the lathe family and is the most widely used. Like the speed lathe, the engine lathe has all the basic parts, e.g., bed, headstock, and tailstock. But its headstock is much more robust in construction and contains an additional mechanism for driving the lathe spindle at multiple speeds. An engine lathe is shown in Fig. Unlike the speed lathe, the engine lathe can feed the cutting tool both in cross and longitudinal direction with reference to the lathe axis with the help of a carriage, feed rod and lead screw. Centre lathes or engine lathes are classified according to methods of transmitting power to the machine. The power may be transmitted by means of a belt, electric motor or through gears.

The bed of the lathe machine needs high compressive strength and vibration damping. Hence, cast iron is used to make the bed given its high carbon content. The spindle of the lathe machine is run by a belt drive, given that there is substantial distance between the headstock and the source of power. The gearbox inside the lathe machine is used to control relative speed between the chuck and the lead screw. The lead screw allows the workpiece to move the carriage in time as the spindle rotates. Controlling the relative speed of the lead screw and the spindle is important for thread cutting.

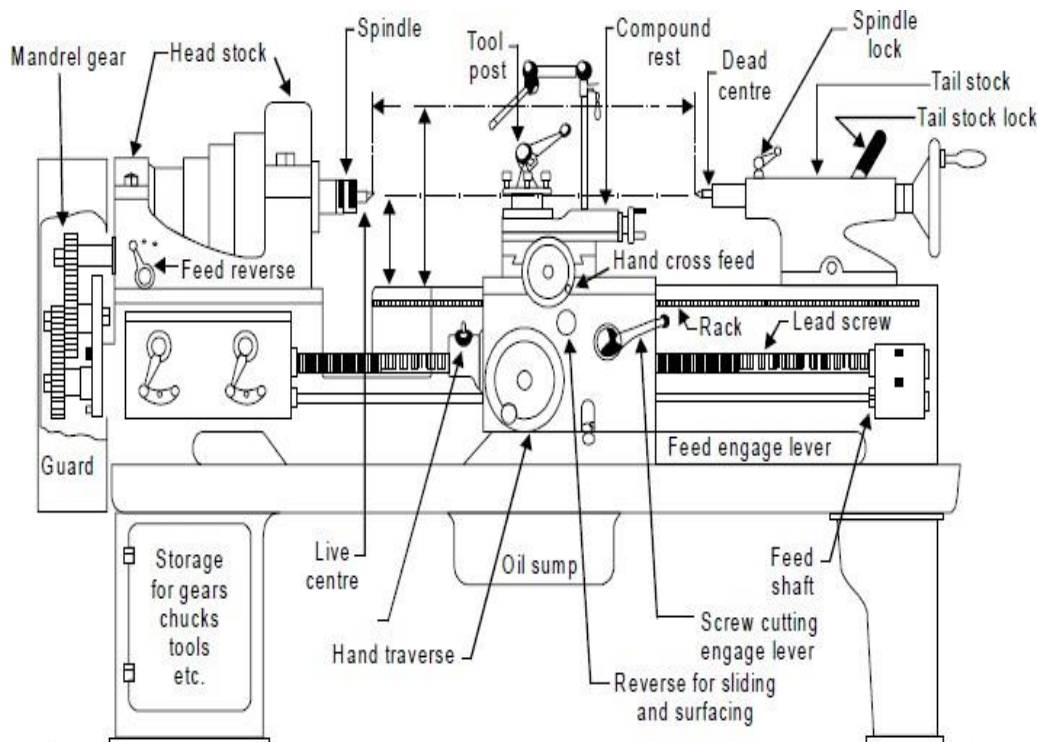


Figure 1: Different parts of engine lathe or Centre lathe

LATHE OPERATIONS

1. Straight turning
2. Taper turning
3. Facing
4. Chamfering
5. Thread cutting
6. Grooving
7. Knurling
8. Under-cutting
9. Filing
10. Spinning
11. Forming
12. Polishing
13. Solder turning
14. Spring winding
15. Boring
16. Drilling

Straight Turning:

Work piece become cylindrical. The motion of the tool is parallel to the work piece surface.

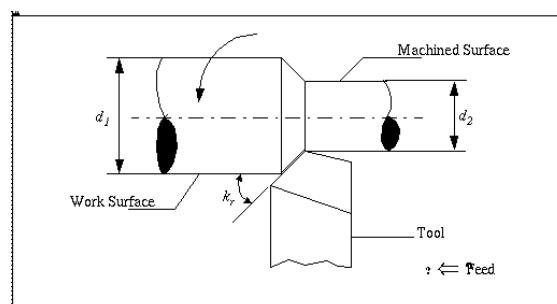
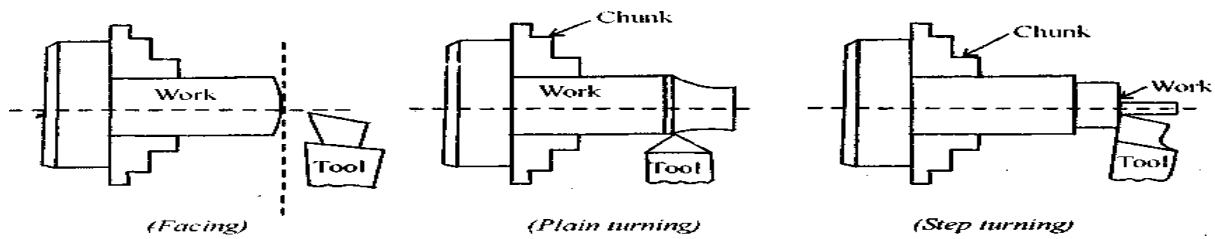


Figure 2: Straight Turning

Facing:

To make side surface perpendicular via cutting tool. The motion of tool is perpendicular to the work piece surface.



Taper Turning:

Dia. of cylindrical work piece decreasing or increasing gradually is called taper turning..

Types of tapers turning:

- By using form tool.
- By setting over tailstock Centre.
- By swiveling the compound rest.
- By using taper turning attachment.

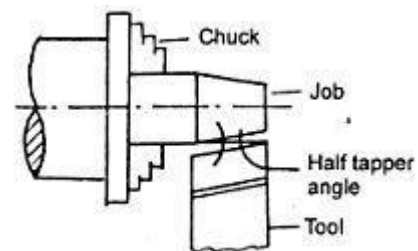
Taper Turning by using form tool:

A broad nose form tool having straight cutting edge makes half taper angle with the axis of work.

The tool is fed right angle to the work axis.

Work is held in chuck or face plate.

Use to turn short length of taper only.



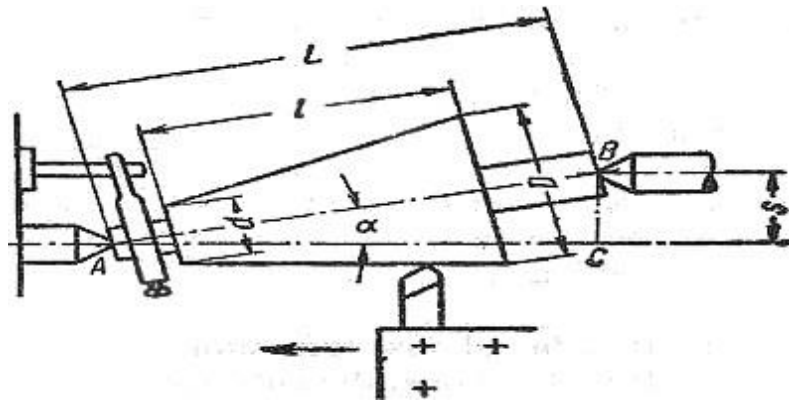
Taper Turning by setting over tail stock:

The method is suitable when the work is held between the centers.

The work pieces are rotated at an angle to the lathe axis and tool fed parallel to the lathe axis.

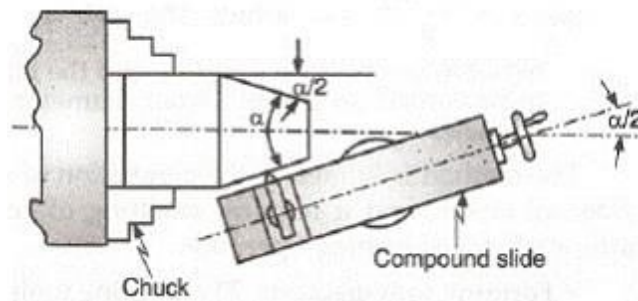
Desired conical surface obtained.

Use to turn small external taper into long work piece.



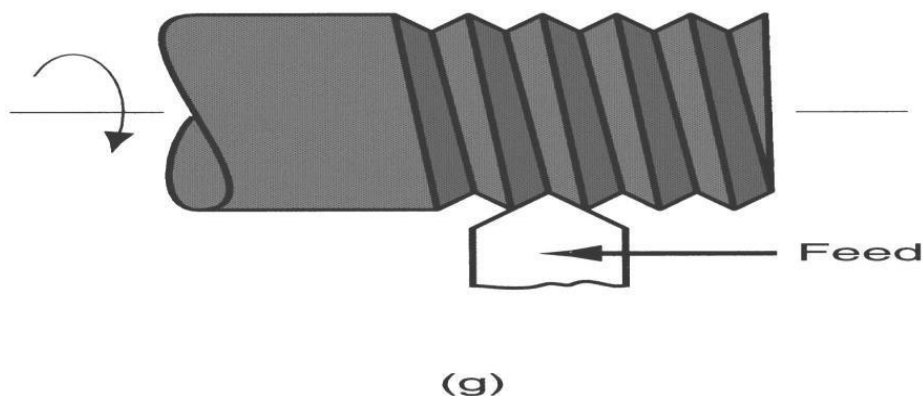
Taper Turning by swiveling the compound rest:

The work piece is rotated at lathe axis and tool is fed at an angle to the axis of rotation of workpiece. Tool is mounted on compound rest. The tool can be fed at angle of compound slide as compound rest is mount at half taper angle.



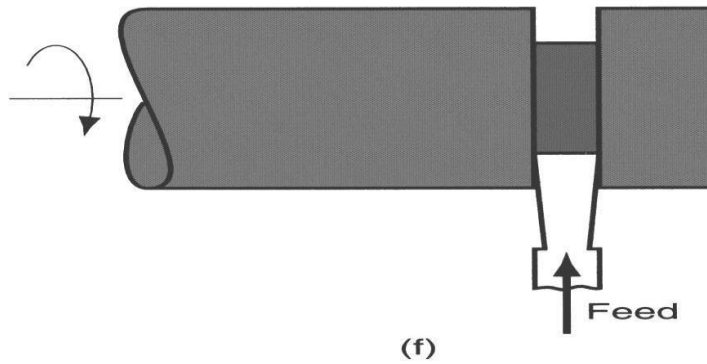
Threading:

Pointed form tool is fed linearly across surface of rotating work part parallel to axis of rotation at a large feed rate, thus creating threads.



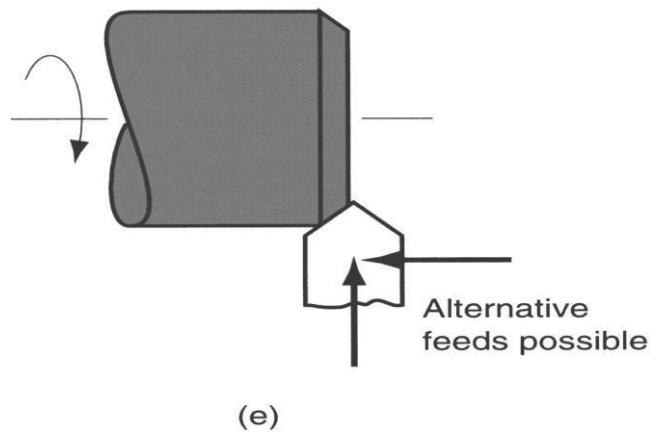
Grooving:

Tool is fed radially into rotating work at some location to cut off end of part or provide a groove.



Chamfering:

Cutting edge cuts an angle on the corner of the cylinder, forming a "chamfer". It removes sharp edges, removes edge irregularities and enables easy assembly in holes.



Boring:

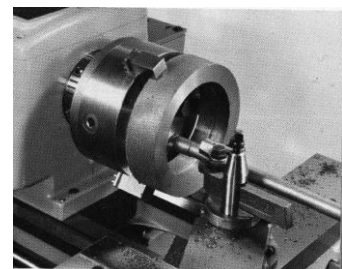
Boring produces circular internal profiles in hollow work pieces

Boring mills are used for large work pieces.

Holes can be bored up to 20M if needed.

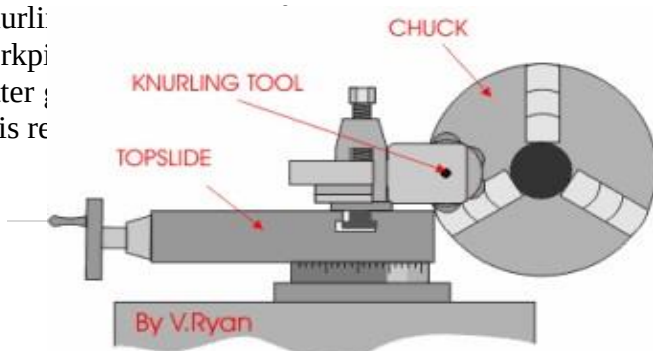
Machines are available with a variety of features:-

Horizontal boring machines.



Knurling:

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Single Point Cutting Tool Definition

The single point cutting tool consists of a sharpened cutting part called its point and the shank. The point of the tool is bounded by the face (along which the chips slide as they are cut by the tool), the side flank or major flank the end flank or minor flank, and the base. As we know we perform several operations on the lathe (like turning and facing) from the single point cutting tool. The tool is usually made of high-speed steel, which has a higher carbon content than the workpiece material, mild steel. Cast iron is not used as a tool material despite its high carbon content as it is brittle.

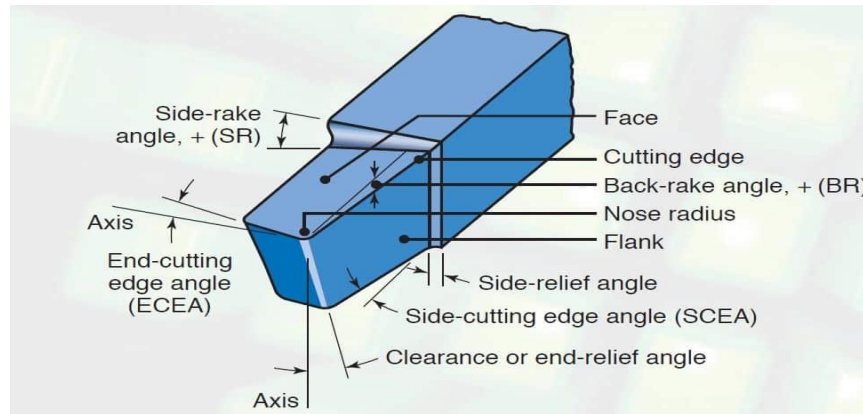


Figure 3: Single point cutting tool.

Types: There are only two types of tools.

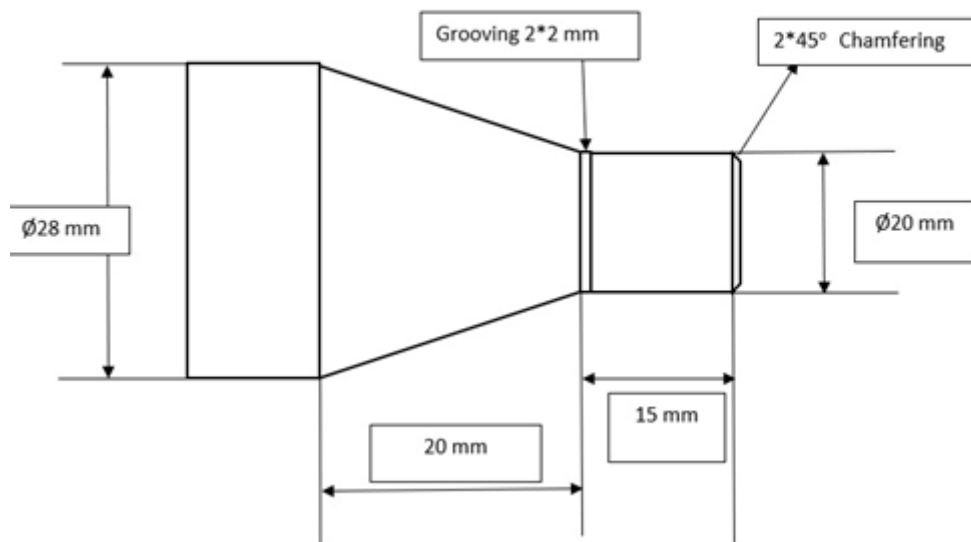
Single point cutting tool.

Multi-Point cutting tool.

Single Point Cutting Tool: Lathe Machine, Planning Machine tool etc.

Multi-Point Cutting Tool: Milling cutter, Grinding wheel, drill tool, broaching, etc.

Machine Shop Job Description



Job type: Taper mandrel

Sequence of Operation:

1. TRUEING (Centering)
2. TOOL CENTERING
3. FACING
4. TURNING
5. MARKING A LINE OF 15mm
6. CHAMFERING

Cutting tool material: HSS (High Speed Steel)

Composition: 18:4:1 (Tungsten: Chromium: Vanadium)

Job Material: Mild Steel, Carbon: 0.25% to 0.35%

Safety precautions

1. Learn how to operate a machine before using the machine.
2. Students are advised not to wear chappals/sandals but to be in shoes.
3. No adjustment, change in speed, repair or oiling should be attempted while the machine is running.

4. Adjustment and security tools, fasten the workpiece in vice and chuck properly.
5. If you are not sure of the correct & safe method of procedure, ask your instructor.
6. Don't touch any moving part or cutting edge of the tool when machining is in progress.
7. Machine should not be left unattended while machine is running.
8. Always wear tight clothes, as loose clothes are likely to be caught in a machine.
9. While working on Grinding machines, always wear goggles.
10. When you have finished using the machine release all automatic levers and clean it.

Questions (Descriptive)

Q1	Define Step-Turning?
Ans	Step turning is an operation performed on lathe machine where the excess material is removed from the workpiece to obtain various steps of different diameters.

Q2	What is a lathe machine?
Ans	A lathe is a machine tool which rotates the workpiece on its axis to perform various operations such as cutting, sanding, knurling, drilling, or deformation, facing, turning, with tools that are applied to the workpiece to create an object which has symmetry about an axis of rotation.

Q3	What are the common machines found in a machine shop?
Ans	The common machines in a machine shop are a lathe, a drilling m/c, a milling machine, a shaper, planer, grinding machine etc.

Q4	Different tool materials used as a cutting tool in Turning process?
Ans	1. HSS tools 3. Diamond tools 2. Ceramic tool 4. Carbide tools

Q5	Different operations performed on a lathe machine?
Ans	1. Straight Turning 2. Step Turning 3. Boring 4. Threading 5. Knurling 6. Facing 7. Parting 8. Drilling 9. Reaming 10. Spinning

Q6.	Name five major parts of lathe machine?
Ans	(i) Bed (ii) Tail stock (iii) Carriage (iv) Chuck (v) Tool post

Q7.	What is the difference between holding methods of three jaw chuck and four jaw chuck?
Ans	Three jaw chuck is self-centering chuck whereas four jaw chuck is independent chuck i.e., each jaw is independently moved.

Q8.	Define Depth of Cut in milling operations.
Ans	The depth of cut in milling is defined as the thickness of the material removed in one pass of the work under the cutter. Thus, it is the perpendicular distance measured between the original and final surface of the workpiece, and is expressed in mm.

Q9.	Write the name of five types of milling cutters.
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Ans	1. Plain milling cutter 2. Side milling cutter 3. Face milling cutter 4. Angle milling cutter. 5. End milling cutter
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Q.10	Describe the need of the operation called knurling.
Ans	Knurling allows hands or fingers to get a better grip on the knurled object than would be provided by the originally smooth metal surface.

Q11.	Why 3-jaw chuck is called self-centered chuck?
Ans	In a 3-jaw chuck, all the jaws are tightened by only moving of single jaw. All the 3 jaws are moved by the same distance. These chucks are very much used for cylindrical jobs.

Q12.	What is the primary type of cutting operation performed in a lathe machine?
Ans	Turning

Q13.	Which material is the lathe machine bed made of and why?
Ans	Cast iron, Because of its high carbon content, it has high compressive strength and vibration damping capacity.

Q14.	What is the function of the gearbox in the lathe machine?
Ans	To control the relative speed between the chuck and the lead screw.

Q15.	What does the lead screw do during thread cutting?
Ans	Allows us to move the workpiece automatically using the power feed
Q16.	What is the drive mechanism of the drilling machine?

Ans	Rack and pinion mechanism: when the hand wheel is rotated, it converts the rotational motion to linear motion by using rack and pinion.
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Q17.	Why is a V-belt used instead of a flat belt between the stepped pulleys for power transmission in a drilling machine?
Ans	To increase the power efficiency

Q18.	Why is a high material removal rate not attempted during grinding?
Ans	Grinding is a finishing operation and hence to maintain a good surface finish, a low material removal rate is desired.

Q19.	What are the key functions of coolants in machining processes?
Ans	To reduce friction, To remove chips and prevent these from damaging the workpiece, To improve tool life by easing material removal, To dissipate heat, To reduce wear and tear of the tool

Q20.	What are the commonly used abrasives for grinding?
Ans	1. CBN (Cubic Boron Nitride) 2. Aluminum Oxide 3. Diamond 4. SiC (Silicon Carbide)

Q21.	Gears are popularly cut on which types of machines?
Ans	Milling Machines

Q22.	What types of motion do the cutter and the workpiece have in a milling operation?
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Ans	Cutter: Rotary Workpiece: Linear
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Q23.	What types of motion do the cutter and the workpiece have in a turning operation?
Ans	Cutter: Linear Workpiece: Rotary

Q24.	What is the difference between the motions of the cutter and the workpiece in a shaping and in a planing operation?
Ans	Shaping: The tool reciprocates, and workpiece is held stationary Planing: Workpiece reciprocates, and tool is stationary

Q25.	Why is chamfering done?
Ans	To remove sharp edges, thus preventing injuries. To remove edge irregularities; easy assembly in holes.

Q26.	which type of power drive is used drive the spindle of the lathe and why?
Ans	Belt Drive, Due to substantial distance from the power source and the head stock

Q27.	What is the difference between single point cutting tool and multi-point cutting tool?
Ans	The major difference between single and multi-point cutting tools is that single point cutting tools have only one main cutting edge, whereas multi-point cutting tools have multiple cutting edges.
Q28.	What is the job that you have prepared on the lathe machine? Describe its utility in machining.
Ans	The job that we have prepared on the lathe machine is a taper mandrel. Taper mandrels are used to hold workpieces (say while holding partly machined workpieces for the purpose of finishing) or as a core for material to be bent, forged or molded. The taper enables a tight frictional grip if the mandrel is pressed into a previously machined hole to act as a support during machining. If the mandrel is

	used to carry tools, then it is known as an Arbor.
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Questions (Objective)

Sl. no.	Questions	A	B	C	D
1	Carriage consists of	Drawslide	Compound slide	Tool post	All of these
2	Which is a self-centering chuck	3 Jawchuck	4 Jawchuck	Independent jaw chuck	None of these
3	Step turning is used to	Reduce the diameter	Increase the diameter	Reduce the length	Making joints
4	Tool to measure Outer diameter	Inside caliper	Vice	Outside caliper	Chisel
5	Drilling operation can be performed on	Both drilling machine and lathe	Only on lathe	Only on Drilling machine	Neither of these two
6	Which one of the following materials is used for manufacturing cutting tools?	Cast steel	Mild steel	High speed steel	Cast iron
7	The commonly used proportions of alloying elements in HSS are	18% tungsten, 4% chromium and 1% vanadium	18% chromium, 4% tungsten and 1% vanadium	18% vanadium, 4% chromium and 1% tungsten	18% cobalt, 4% chromium and 1% vanadium
8	Which one of the following standard tapers is self-holding type?	Morse taper	Brown and Sharpe taper	Jarno taper	Metric taper
9	An operation of embossing, producing a roughened diamond-shaped pattern on a smooth surface of a cylindrical job to provide effective gripping is known as:	Grooving	Knurling	Turning	All of the above

10	Tool used in turning operation	Singlepoint cuttingtool	Cutter	Multipoint cutting tool	Metal Remover
11	Bed of lathe is generally made by:	Cast iron	Mild steel	High speed steel	None
12	The non-circular jobs can be suitably held in:	three jaw chuck	Four jaw Chuck	Collet Chuck	Mandrel
13	The part of a lathe set on the carriage that carries the tool post and holder is known as:	Tool head	Tail Stock	Compound Rest	Cross Slide
14	Feed motion is provided by the cutting toolin:	Lathe and Shaper	Lathe and Drill Machine	Shaper and milling machine	Milling and drilling machine
15	Flat surface can be produced in:	Shaping machine only	Planning machine only	Slotting machine only	All the above

16	Facing operation produces what type ofshape?	Tapered	Flat	Curved	Helical
17	Which operation is done to have a better gripon the job surface?	Facing	Turning	Knurling	Chamfering
18	Sharp circular edges are removed by?	Facing	Turning	Knurling	Chamfering
19	Turning reduces which of these in a job?	Diameter	Length	Curved surface area	None
20	Tailstock and dead Centre helps in?	Tool centering	Job centering	Accommodating longer length jobs	All
21	Tool used in turning operation	Cutter	Multipoint cutting tool	Single point cutting tool	Metal Remover
22	The carriage of the lathe travels along whichaxis:	X axis	Z axis	Y axis	none ofthese

Chapter 2

Sheet Metal Shop

Introduction

Sheet metal is a versatile material used in a wide range of industries, including automotive, aerospace, construction, and manufacturing. It is characterized by its thin, flat structure, making it ideal for forming various shapes and structures. In the Sheet Metal Shop, you will learn how to transform raw sheet metal into functional components, such as ductwork, enclosures, brackets, and more.

The skills acquired in the Sheet Metal Shop are not only valuable in industrial settings but can also be applied in personal projects and hobbies. Whether you aspire to become a professional sheet metal worker, pursue a career in engineering, or simply wish to explore the art of metalworking, this manual will serve as an essential resource.

It is important to note that sheet metal fabrication requires attention to detail, precision, and patience. It is a craft that combines technical knowledge with hands-on skills, allowing you to bring your ideas to life in three dimensions.

Machines and Operations:

Ball/Screw Press (For light press work): The ball/screw press is a machine used for light pressing operations. It utilizes a ball or screw mechanism to apply pressure to the workpiece. This machine is suitable for applications that require moderate force, such as assembling components or pressing small parts together.

Bending Machine (For sheet bending up to 45 degrees): A bending machine is designed to bend sheet metal up to a maximum angle of 45 degrees. It employs a bending die and a clamping mechanism to hold and shape the sheet metal. This machine is commonly used in metal fabrication for creating angular bends in sheet metal components.

Bending Rollers (For piping, bending, curving): Bending rollers, also known as roll benders or section rollers, are used for bending and curving various materials, including sheet metal and piping. This machine consists of three rollers that apply pressure to the workpiece, allowing it to be formed into curved or circular shapes. Bending rollers are commonly used in the fabrication of pipes, tubes, and cylindrical components.

Double Action Hydraulic Press (For sheet forming operations): The double action hydraulic press is a versatile machine used for various sheet metal forming operations. It utilizes hydraulic power to apply force on the workpiece from both the top and bottom directions. This allows for precise and efficient shaping of sheet metal into desired forms, such as deep drawing, embossing, and coining.

Folding Machine (For sheet bending up to 90 degrees): A folding machine, also known as a press brake, is designed to bend sheet metal up to a maximum angle of 90 degrees. It uses a punch and die set to apply controlled pressure and create precise bends in the sheet metal. Folding machines are widely used in sheet metal fabrication for creating box-shaped components, channels, and flanges.

High Temperature Furnace (For heat treatment): A high-temperature furnace is used for heat treatment processes in the sheet metal industry. It provides a controlled environment with elevated temperatures to alter the physical properties of the metal, such as hardness, strength, and ductility. Heat treatment processes include annealing, tempering, quenching, and stress relieving.

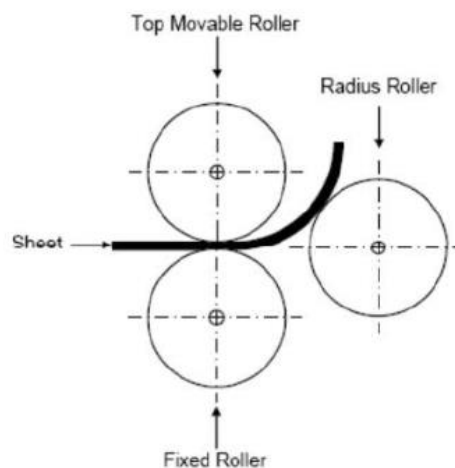
Lathe (For spinning sheet metal parts): A lathe is a versatile machine tool used for various operations, including spinning sheet metal parts. In sheet metal spinning, the metal sheet is clamped to a rotating mandrel while a forming tool applies pressure to shape the sheet into a desired form, such as a cylindrical or conical shape. Lathes are widely used in metalworking and sheet metal fabrication for creating precision components.

Manual Shearing Machine (For sheet cutting up to 1mm thickness): A manual shearing machine, also known as a hand shear, is used for cutting sheet metal up to a maximum thickness of 1mm. It consists of a fixed blade and a movable blade that shear the sheet metal as it is pressed between them. Manual shearing machines are commonly used for straight cuts and trimming of thin sheet metal.

Mechanical Shearing Press (For cutting tensile specimens): A mechanical shearing press is specifically designed for cutting tensile specimens from sheet metal. It utilizes a shearing mechanism to precisely cut the metal into required shapes and sizes for mechanical testing purposes. These machines are commonly used in material testing laboratories and research facilities.

Nibbling Machine (For beading): A nibbling machine is used for beading operations in sheet metal fabrication. It employs a punch and die set to create intricate shapes or patterns, such as beads or flutes, on the sheet metal surface. Nibbling machines are commonly used in automotive, aerospace, and architectural industries for decorative or functional purposes.

Rolling Mill (For cold rolling of sheets and strips): A rolling mill is a machine used for the cold rolling of sheets and strips of metal. It consists of a series of rollers that compress and shape the metal as it passes through them. Cold rolling is a process that reduces the thickness and refines the surface finish of the metal while maintaining its strength. Rolling mills are commonly used in industries such as manufacturing, construction, and automotive for producing thin sheets, strips, and coils of metal with precise dimensions and surface characteristics.



Shearing Machine (For sheet cutting up to 3 mm thickness): A shearing machine is a powerful machine used for cutting sheet metal up to a maximum thickness of 3 mm. It employs a shear mechanism that applies a shearing force to the metal, causing it to fracture and separate along a straight cutting line. Shearing machines can be operated manually, mechanically, or hydraulically, depending on the size and thickness of the metal being cut. They are widely used in metal fabrication workshops, construction sites, and manufacturing plants for cutting sheets of metal into desired sizes and shapes.

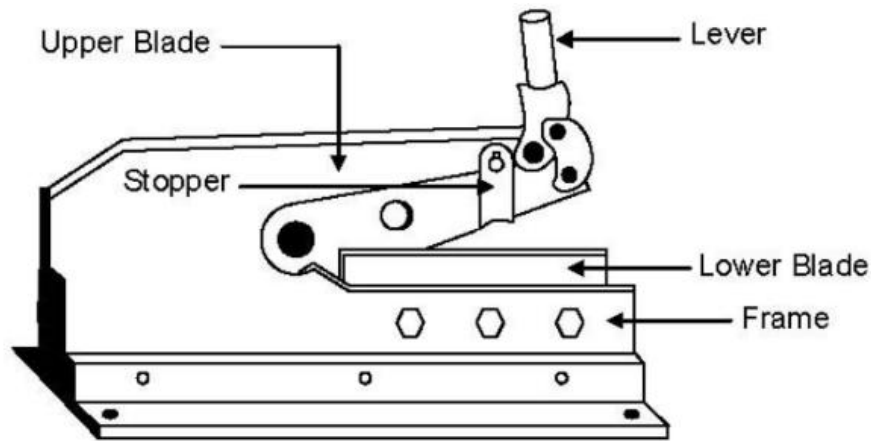


Figure: Manual shearing machine

Vacuum Forming Machine (For thermoforming of plastics): A vacuum forming machine is used for thermoforming plastics to create three-dimensional shapes. It involves heating a plastic sheet until it becomes pliable, placing it over a mould, and applying a vacuum to draw the sheet onto the mould's contours. The vacuum creates suction, causing the heated plastic to conform to the mould's shape. Once cooled, the plastic retains the desired shape. Vacuum forming machines are commonly used in industries such as packaging, automotive, and signage for producing plastic components, trays, and packaging materials with precise shapes and sizes.

Materials used in sheet metal work.

(a) Black iron sheet

It has a bluish -black appearance and is often referred to as uncoated sheet. Since it is uncoated, therefore, it corrodes rapidly.

(b) Galvanized iron

It is soft steel coated with molten zinc. The zinc coating resists rust, improves the appearance of the metal, and permits it to be soldered with greater ease. This sheet is used in fabricated products such as pans, buckets, furnaces, heating ducts, cabinets, gutters and in many other articles.

(c) Copper

It is a reddish coloured metal and is extremely malleable and ductile. It is used extensively in the electrical field.

(d) Aluminium

It is silvery white coloured metal and has many qualities like high ratio of strength to weight, corrosion resistant qualities and ease in fabrication.

(e) Stainless Steel

It is an alloy steel possessing the ability to resist corrosion without any surface coating. One important type of stainless steel contains 18% chromium and 8% nickel. It is widely used in building streamlined trains, food handling equipment, kitchen wares and in many other applications which require great strength and resistant to corrosion.

Tools and Equipment

The following tools and equipment are commonly used in sheet metal shop:

Hand tools: Hammers, pliers, wrenches, screwdrivers, files, saws, and drills

Power tools: Shears, brakes, punches, and drills

Measuring and marking tools: Rulers, callipers, squares, and protractors

Tools:

Centre Punch: The centre punch is similar to prick punch, but its point is ground to an angle of approximately 90° . It is used to mark the location of bend lines on heavy metal and to mark the centres of holes to be drilled.

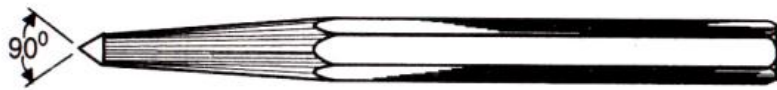


Figure: Centre Punch

Hammer: The ball peen hammer has a round, slightly curved face and a round head. It is general purpose hammer.

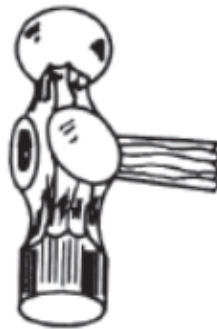


Figure: Hammer

Mallet: The mallets may be made from hide, fibre or wood. The best size of mallet is 5-10 cm diameter. These may be obtained in various shapes to suit special work. It is used instead of hammer to avoid dents in sheet metal.



Figure: Mallet

Flat nose plier: The flat nose plier has flat jaws with small grooves. It is used for forming and holding work.

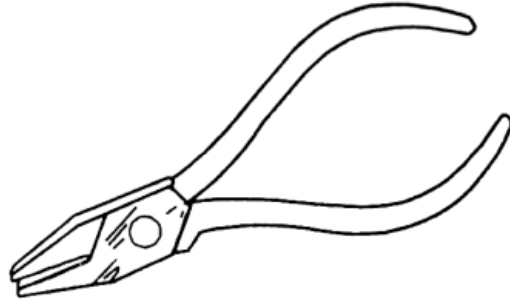


Figure: Flat nose plier

Round nose plier: The round nose plier has long jaws rounded on the outside. It is used for holding and forming the various shapes and patterns.

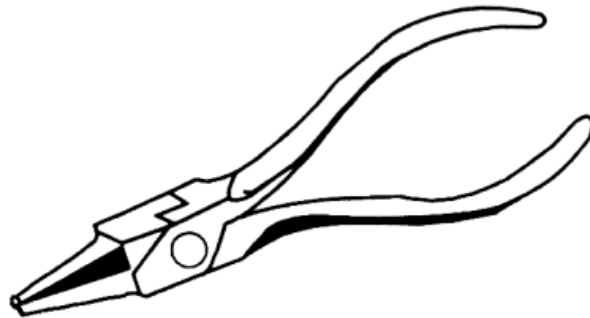


Figure: Round nose plier

Funnel Stake: The funnel stake is used for planishing tapered work and hand forming of funnels and similar conical shapes of sheet metal.

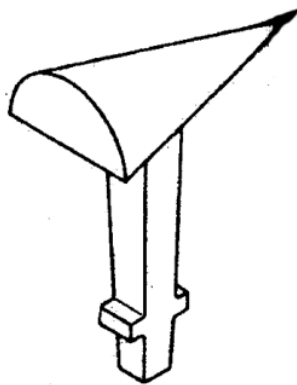


Figure: Funnel Stake

Beak Horn Stake: The beak horn stake has a round tapered horn at one end and a rectangular tapered horn at the other end. It is basically used for forming, riveting, and seaming articles made of sheet metal parts. It is not much suitable like blowhorn stake.

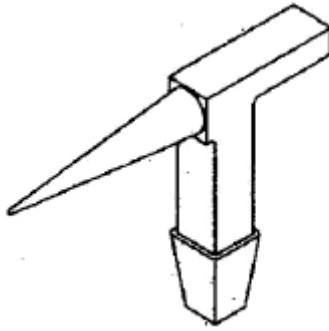


Figure: Beak Horn Stake

Creasing Stake: This has two rectangular shaped horns. One is plane and the other contains a series of grooved slots of various sizes. These slots are used for wiring ‘sinking’ a bead on the straight edges of the flat sheet, and for making small diameter tubes with thin gauge metal.

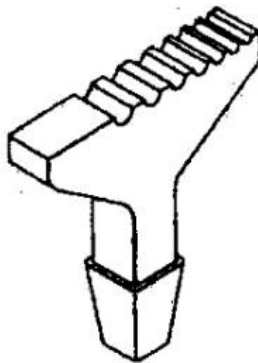


Figure: Creasing Stake

Lab Operations:

Sheet Operations:

Measurement and marking

Cutting

Bending

Joining

Finishing

Joints Operation:

Hemming: In this process sheets are joined by bending it usually to 180°. Automotive body panels and automotive parts such as deck-lids, trunk-lids, doors, hoods and tailgates are formed by hemming process.



Figure: Single and double Hem

Locked Groove Joint: It is self-locking joint used commonly in the fabrication of cylindrical and conical objects.



Figure: Locked Groove Joint

Knocked-up Joint (Corner Joint): In this joint, the ends of the two pieces of sheet are cut at a 45-degree angle, creating a metered corner. When assembled, the two pieces fit together to form a right angle. The joint is then secured with glue, nails, screws, or other fasteners, depending on the specific application.



Figure: Corner Joint

Flanging: In this operation the edge of the metal is manipulated to create a flange, which is a raised, folded, or curved portion along the edge. This flange can serve various purposes, such as providing stiffness, increasing strength, improving alignment, or creating a mating surface for joining with other components.

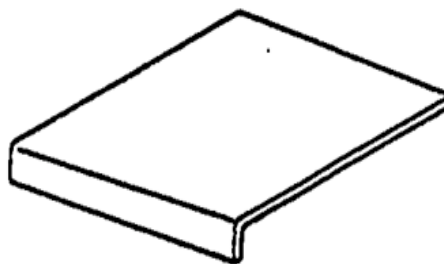


Figure: Flanging

Sheet Metal Workshop Practical

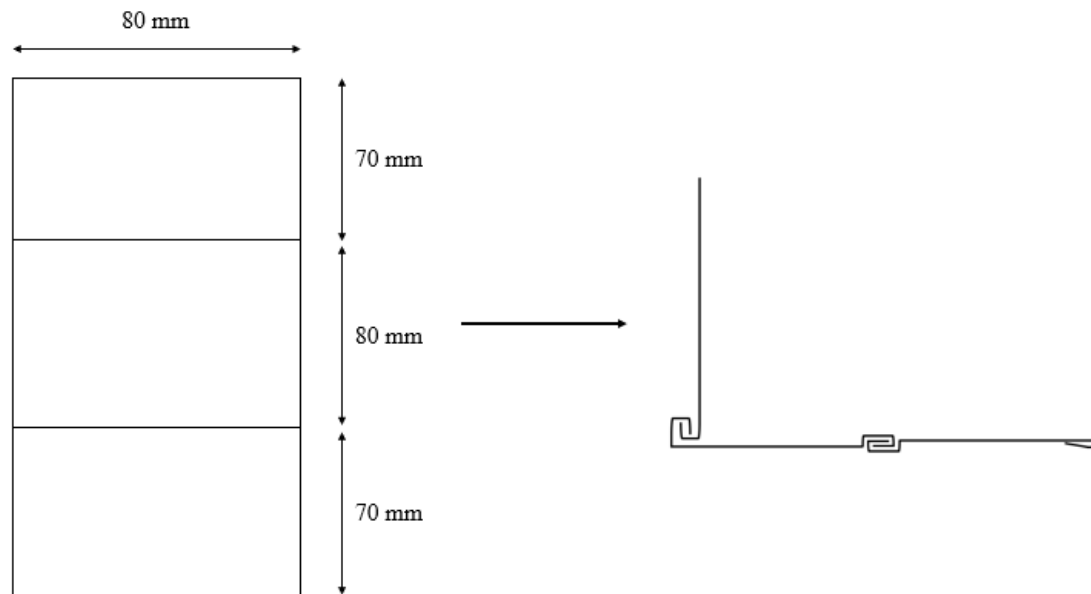


Figure: Sheet metal job to be performed

Safety Precautions:

Wear Proper Personal Protective Equipment (PPE): Always wear the appropriate PPE before entering the sheet metal shop. This includes safety glasses, gloves, ear protection, and a protective apron. Additionally, make sure to wear sturdy closed-toe shoes to protect your feet from potential hazards.

Secure the Workpiece and Machine: Before starting any operation, ensure that the sheet metal and the machine tools are securely clamped or fastened in place. This will prevent unintended movements or accidents during the fabrication process.

Follow Machine Operating Procedures: Familiarize yourself with the specific operating procedures for each machine tool in the sheet metal shop. Adhere to the manufacturer's instructions and guidelines when setting up, operating, and adjusting the machines.

Be Aware of Sharp Edges and Moving Parts: Sheet metal often has sharp edges that can cause cuts or injuries. Take caution when handling and working with sheet metal to avoid accidental contact with these edges. Additionally, be mindful of moving parts on the machine tools and keep your hands and fingers clear of the cutting, bending, or punching areas.

Use Caution with Power Tools: Some sheet metal operations involve the use of power tools such as shearing machines or press brakes. Exercise caution and maintain a firm grip on the tools when operating them. Avoid distractions and focus on the task at hand to prevent accidents.

Keep the Work Area Clean: Maintain a clean and organized work area. Remove any unnecessary tools, debris, or tripping hazards from the vicinity. A clean workspace promotes better visibility, reduces the risk of accidents, and allows for more efficient work.

Seek Proper Training: Before using any machine tool or performing sheet metal operations, ensure that you have received proper training and guidance from a knowledgeable instructor or supervisor.

Understanding the correct techniques and safety protocols is crucial for safe and effective work.

Report Any Safety Concerns: If you notice any safety hazards, malfunctioning equipment, or potential risks in the sheet metal shop, promptly report them to a supervisor or designated authority. It is essential to maintain a safe working environment for everyone involved.

Remember, your safety is of utmost importance. By following these safety precautions and guidelines, you can ensure a secure and productive experience in the sheet metal shop.

Basic Skills Required:

The following are some basic skills that students will learn in sheet metal shop:

Marking: Students will learn how to use measuring and marking tools to mark sheet metal for cutting and bending.

Cutting: Students will learn how to use hand tools and power tools to cut sheet metal.

Bending: Students will learn how to use hand tools and power tools to bend sheet metal.

Soldering: Students will learn how to solder sheet metal joints.

Welding: Students will learn how to weld sheet metal joints.

Questions and Answers:

1. what is the difference between sheet, foil, and plate?

Ans: Thickness

Sheet = 0.5mm – 6mm

Foil = < 0.2 mm

Plate > 6mm

2. Which sheets are coated with zinc to prevent rust formation?

Ans: Galvanized iron sheet

3. How do you join sheet metal?

Ans: riveting, spot welding, and seam welding.

4. To make a small indentation (centre hole) in sheet metal, a _____ is used.

Ans: Centre Punch

5. To bend the sheet metal by 180 degrees a ————— is used.

Ans: Mallet

6. Files are used to ————— metal sheet surface.

Ans: Finish

7. How much thickness of the sheet can be cut by a manual sharing machine and power shearing machine?

Ans: manual sharing machine for soft metal 1.5mm and for hard metal 1mm

Power-sharing machine for soft metal 5 mm and for hard metal 4 mm

8. Which hand tool is used for cutting the sheets?

Ans: Snip tool

9. The formation of rough edges or ridges on the sheet metal piece is called?

Ans: Burr

10. What are the applications of sheet metal?

Ans: automobile and truck bodies, airplane fuselages, and wings, tinplate for tin cans, roofing for buildings, and home appliances.

Q. 11 What is a progressive die?

Ans: A “progressive die” is a series of punch press tools that have been specially designed to perform all the punching, cutting, coining, and bending required to produce a metal part through a multistep process.

Q. 12 Why progressive die are used?

Ans. Progressive die stamping is an excellent manufacturing solution to manufacture complex parts in large numbers.

Q.13 _____ snip is used for trimming cylindrical or conical work?

Ans. Bent snip.

Q.14 What is difference between lancing and slitting?

Ans. In Lancing is creating partial cut in sheet, so that material is left attached to sheet. slitting is cutting straight lines in sheet, such that no scrap material is produced.

Q.15 What is Hemming?

Ans: Hemming is folding the edge of the sheet over itself to increase thickness of part.

Q.16 Difference between mallet and hammer?

Ans. Mallet is used for giving light blows, whereas for heavy blows, hammer is used.

Q.17 Which material is Anvil made of?

Ans. Mild steel or cast steel with hardened surface.

Q.18 what is the use of Chisels?

Ans. When there is no approach or access to line to be cut by snips/shears from an edge the material is cut using chisel.

Q.19 Name of any five measuring instruments?

Ans. (i) Try square (ii) Vernier calliper (iii) Micrometre (iv) Divider (v) Inside calliper.

Q. 20 What is the use of divider?

Ans. It is used for marking arc, dividing a line, or transferring the dimension.

Chapter 3

Fitting Shop

Introduction

Manufacturing processes are broadly classified into four categories: (i) Casting processes, (ii) Forming processes, (iii) Fabrication processes, and (iv) Material removal processes. In all these processes, components are produced with the help of either machines or manual effort. The attention of a fitter is required at various stages of manufacture starting from marking to assembling and testing the finished goods. Working on components with hand tools and instruments, mostly on work benches, is generally referred to as 'Fitting work'.

Machine tools can produce work at a faster rate, but there are occasions when components are processed at the bench. Sometimes, it becomes necessary to replace or repair components which must be fit accurately with another component on reassembly. This involves a certain amount of hand fitting. The assembly of machine tools, jigs, gauges, etc., involves a certain amount of bench work. The accuracy of work done depends upon the experience and skill of the fitter.

The term bench work refers to the production of components by hand on the bench, whereas fitting deals with the assembly of mating parts, through removal of metal, to obtain the required fit. Both the bench work and fitting require the use of a few simple hand tools and considerable manual effort. The operations in the above works consist of filing, chipping, scraping, sawing drilling, and tapping.

Holding Tools

Bench Vice

The bench vice is a work holding device. It is the most used vice in a fitting shop. The bench vice is shown in the figure below.

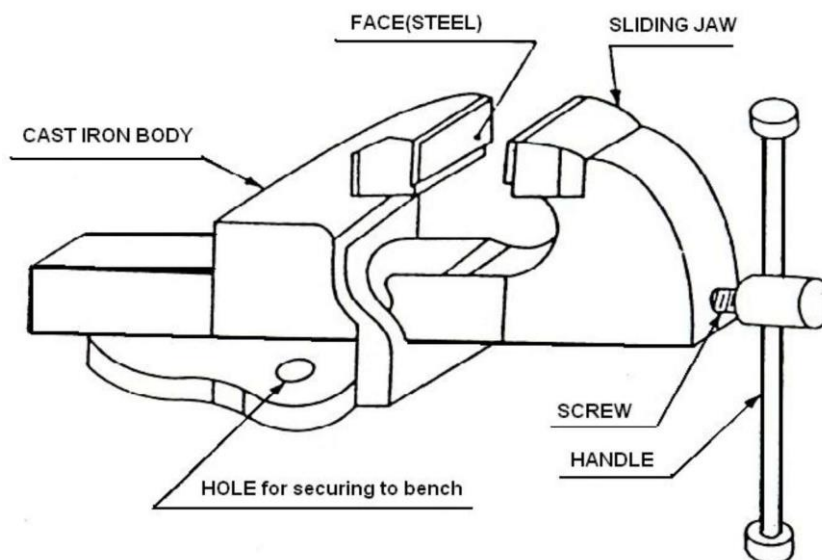


Figure: Bench Vice

It is fixed to the bench with bolts and nuts. The vice body consists of two main parts, fixed jaw and movable jaw. When the vice handle is turned in a clockwise direction, the sliding jaw forces the work against the fixed jaw. Jaw plates are made of hardened steel. Serrations on the jaws ensure a good grip. Jaw caps made of soft material are used to protect finished surfaces, gripped in the vice. The size of the vice is specified by the length of the jaws varying from 75 mm to 300 mm. The vice body is made of cast iron, which is strong in compression, weak in tension and so fractures under shocks and therefore should never be hammered.

V-block with clamp

The V-block is a rectangular or square block with a V-groove on one or both sides, opposite to each other. The angle of the V is usually 90°. V-block with a clamp is used to hold cylindrical work securely, during marking of measurements or for drilling for this the bar is faced longitudinally in the V Groove and the screw of V clamp is tightened. This grip the rod is firm with its axis parallel to the axis of the V-groove.

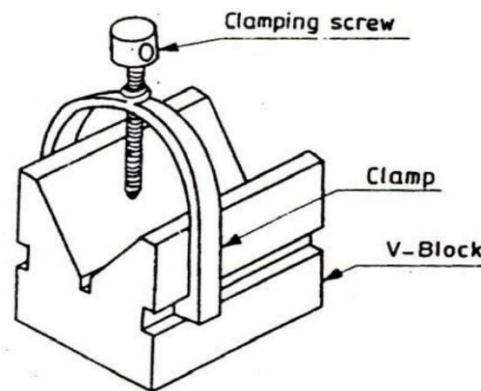


Figure: V- block

C-Clamp

This is used to hold work against an angle plate or block or any other surface when gripping is required. Its fixed jaw is shaped like English alphabet 'C' and the movable jaw is round and directly fitted to the threaded screw at the end. The working principle of this clamp is the same as that of the bench vice.

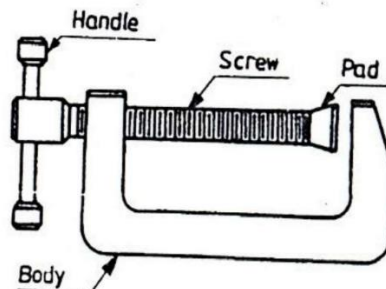


Figure: C clamp

Marking and Measuring Tools

Surface plate

The surface plate is machined to fine limits and is used for testing the flatness of the work piece. It is also used for marking out small boxes and is more precious than the marking table. The degree of the finished depends upon whether it is designed for bench work in a fitting shop or for using in an inspection room; the surface plate is made of Cast Iron, hardened Steel or Granite stone. It is specified by length, width, height, and grade. Handles are provided on two opposite sides, to carry it while shifting from one place to another.

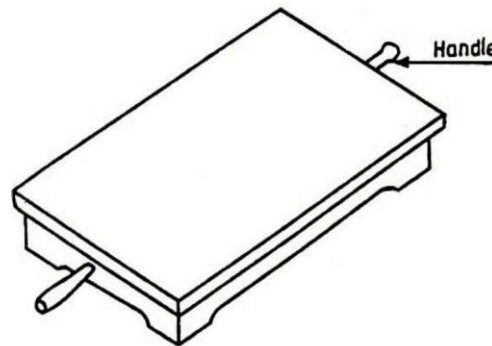


Figure: Surface plate

Angle Plate

The angle plate is made up of cast iron in different sizes. It has two planed surfaces at right angles to each other and has various slots on both surfaces to hold the work firmly by means of bolts and clamps. The angle plate is used with the conjunction of surface plate for supporting work at right angles for marking.

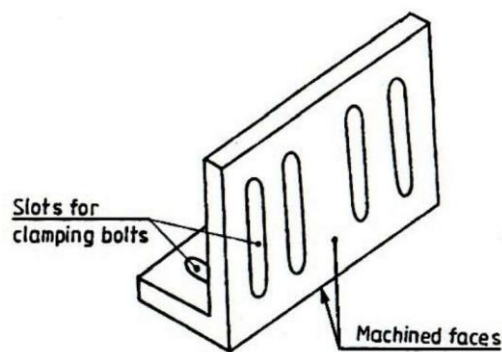


Figure: Angle plate

Try Square

It is used for measuring and marking tool for 90° angle. In practice, it is used for checking the squareness of many types of small works when extreme accuracy is not required. The blade of the Try square is made of hardened steel and the stock of cast Iron or steel. The size of the Try square is specified by the length of the blade.

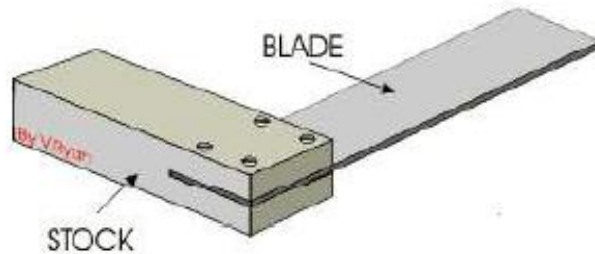


Figure: Try square.

Scriber

A Scriber is a slender steel tool, used to scribe or mark lines on metal work pieces. It is made of hardened and tempered High Carbon Steel. The Tip of the scriber is generally ground at 12° to 15° . It is generally available in lengths, ranging from 125 mm to 250 mm. It has two pointed ends the bent end is used for marking lines where the straight end cannot reach.

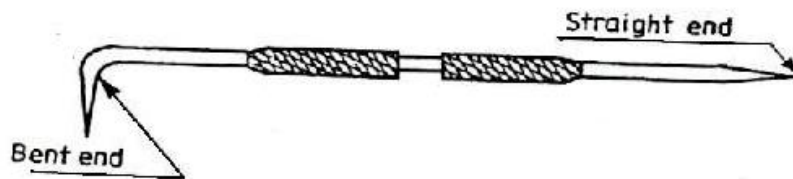


Figure: Scriber

Calipers

They are indirect measuring tools used to measure or transfer linear dimensions. These are used with the help of a steel Rule to check inside and outside measurements. These are made of case-hardened mild steel or hardened and tempered low carbon steel. While using, the legs of the caliper are set against the surface of the work, whether inside or outside and the distance between the legs is measured with the help of a scale and the same can be transferred to another desired place. These are specified by the length of the leg.

In the case of the outside caliper, the legs are bent inwards and in the case of inside caliper, the legs are bent outwards.

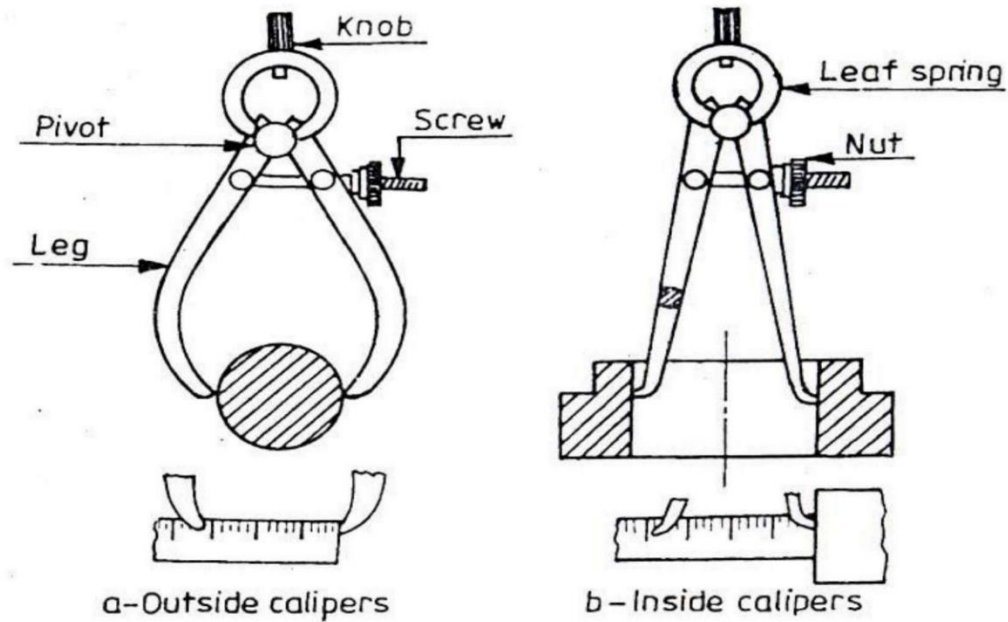


Figure: Calipers

Odd leg Caliper

This is also called ‘Jenny Caliper’ or Hermaphrodite. This is used for marking parallel liners from a finished edge and for locating the center of round bars; it has one leg pointed like a divider and the other leg bent like a caliper. It is specified by the length of the leg up to the hinge point.

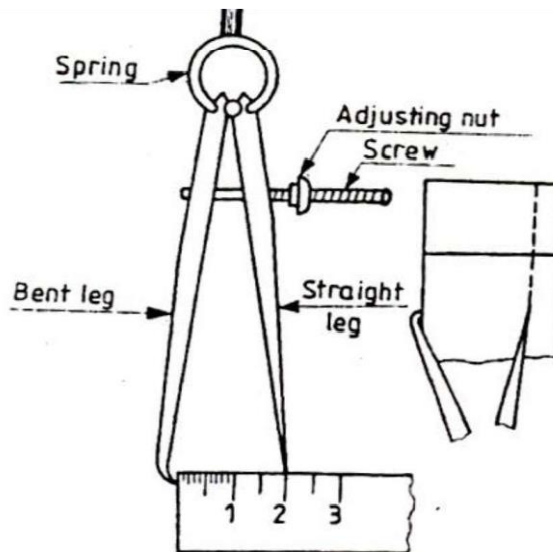


Figure: Odd leg caliper

Divider

It is basically like the calipers except that its legs are kept straight and pointed at the measuring edge. This is used for marking circles, arcs laying out perpendicular lines, by setting lines. It is made of case-hardened mild steel or hardened and tempered low carbon

steel. Its size is specified by the length of the leg.

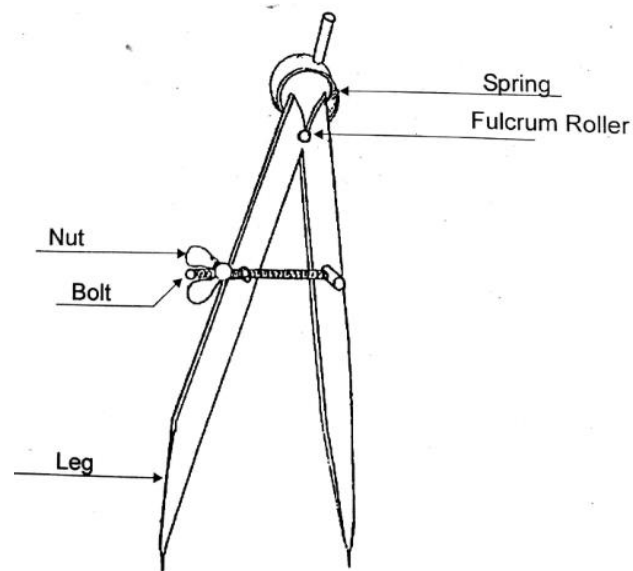


Figure: Divider

Vernier Calipers

These are used for measuring outside as well as inside dimensions accurately. It may also be used as a depth gauge. It has two jaws. One jaw is formed at one end of its main scale and the other jaw is made part of a vernier scale.

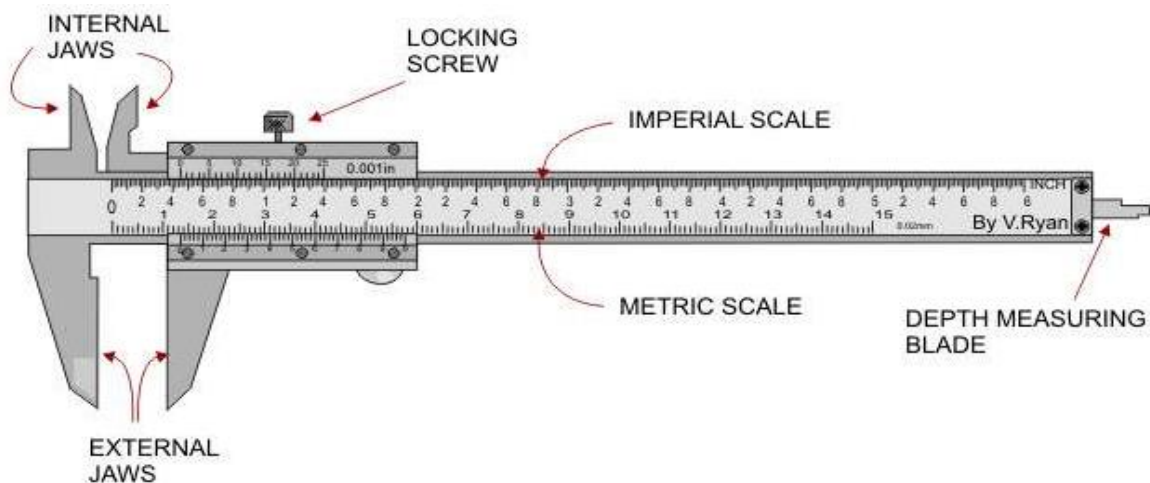


Figure: Vernier Caliper

Outside Micrometer

This is used for measuring external dimensions accurately. Figure shows a micrometer of 0 to 25 mm range with an accuracy of 0.01 mm. These are available in different ranges with interchangeable anvils varying from 0-25 mm to 2000 mm in sizes and 0.01 to 0.001 in accuracy. There are many types of micrometers designed for special purpose use. They include thread micrometers to measure thread dimensions, tube micrometers to measure wall thickness of tubes, etc.

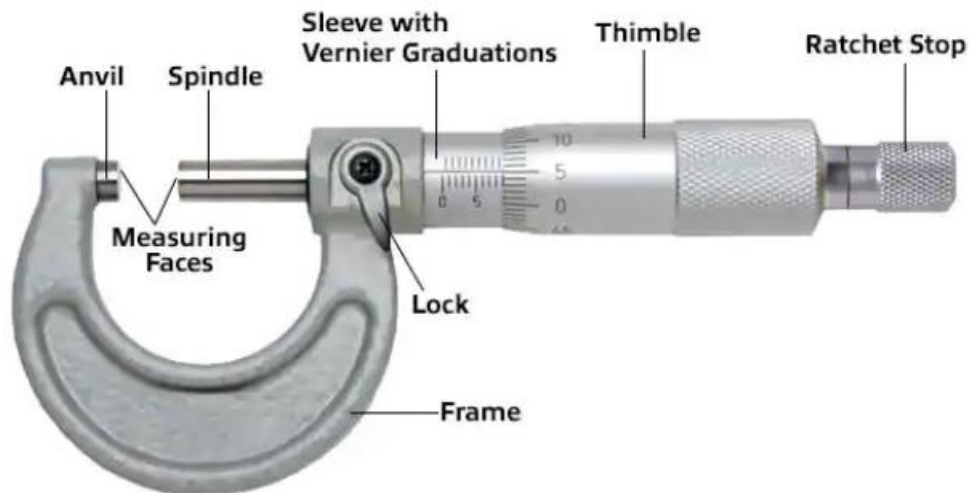


Figure: Outside Micrometer

Inside Micrometer

This is used to measure inside dimensions accurately. Figure shows an inside micrometer. These are available in different ranges and accuracies.



Figure: Inside Micrometer

Punches

A punch is a marking tool used in bench work and is known as center punch. It is generally made from 10 mm octagonal cast steel about 100 mm long. One end of the punch carries a point ground to an included angle of 90° and is tempered to a pale straw color while the other end is slightly chamfered to prevent it from burring. The body is knurled for gripping it by hand. It is used for marking the ends of work to be centered for turning in lathe as well as for all centers of holes for drilling.

The prick or dot punch is like center punch but small. It is made from 6 mm diameter cast steel with a sharp point of 60° . It is used for marking the position of lines and centers of circles to be drawn with the dividers.

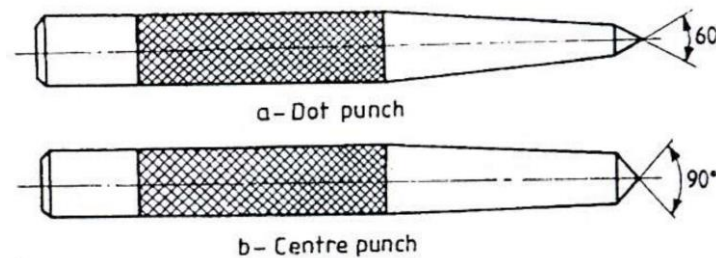


Figure: Punches

Vernier height gauge

The Vernier Height gauge clamped with a scribe. It is used for Lay out work and offset scribe is used when it is required to take measurement from the surface, on which the gauge is standing. The accuracy and working principle of this gauge are the same as those of the vernier calipers. Its size is specified by the maximum height that can be measured by it. It is made of Nickel Chromium Steel.

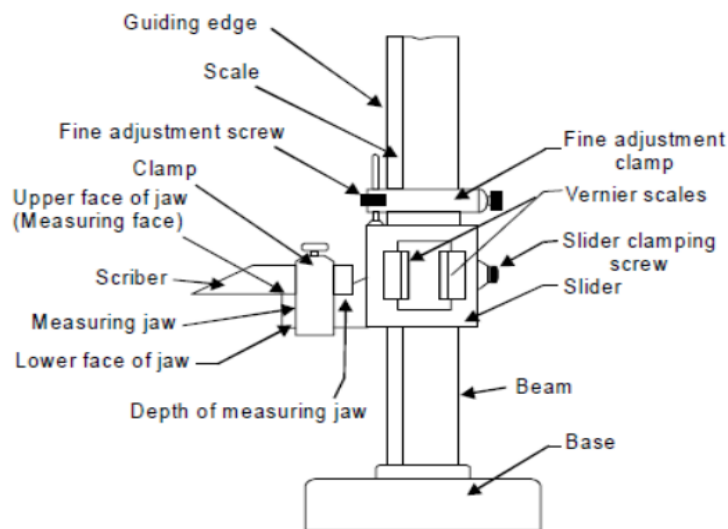


Figure: Vernier height gauge

Feeler Gauge

The thickness gauges or feeler gauges are a set of gauges consisting of thin strips of metal of varying thickness. They are widely used for measuring and checking bearing-clearance, adjusting tappets, spark plug gaps, and so on. The thickness varies from 0.05 to 0.5 mm.

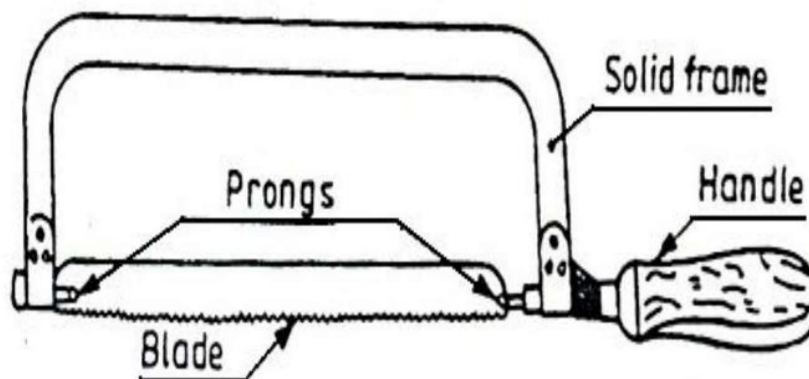


Figure: Feeler gauge

Cutting tools

Hack Saw

The Hack Saw is used for cutting metal by hand. It consists of a frame, which holds a thin blade, firmly in position. A hacksaw blade is specified by the number of teeth per centimeter. Hacksaw blades have several teeth ranging from 5 to 15 per centimeter (cm). Blades having lesser number of teeth per cm are used for cutting soft materials like aluminum, brass, and bronze. Blades having a larger number of teeth per centimeter are used for cutting hard materials like steel and cast Iron. Hacksaw blades are classified as (i) All hard and (ii) flexible type. The all-hard blades are made of H.S.S, hardened and tempered throughout to retain their cutting edge longer. These are used to cut hard metals. These blades are hard and brittle and can break easily by twisting and forcing them into the work while sawing. Flexible blades are made of H.S.S or low alloy steel but only the teeth are hardened, and the rest of the blade is soft and flexible. These are suitable for use by unskilled or semi-skilled people.



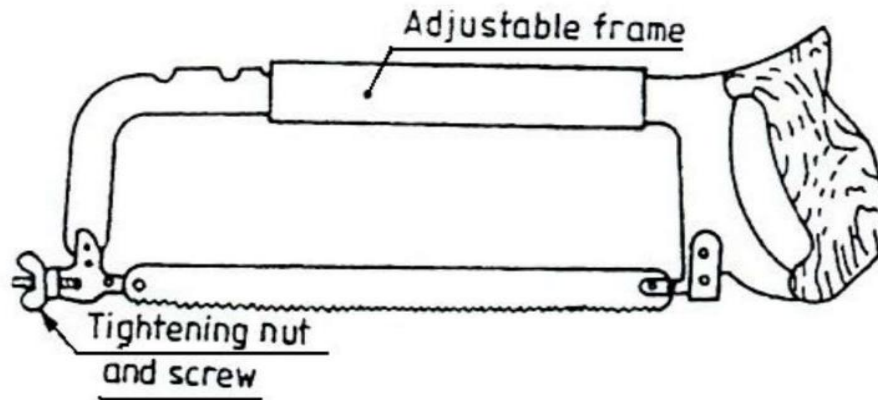


Figure: Hacksaw frame with blade

The teeth of the hacksaw blade are staggered, as shown in figure and known as a “set of teeth”. These make slots wider than the blade thickness, preventing the blade from jamming.



Figure: Blade and slot thickness while in operation

Chisels

Chisels are used for removing surplus metal or for cutting thin sheets. These tools are made from 0.9% to 1.0% carbon steel of octagonal or hexagonal section. Chisels are annealed, hardened, and tempered to produce a tough shank and hard cutting edge. Annealing relieves the internal stresses in a metal. The cutting angle of the chisel for general purpose is about 60°.

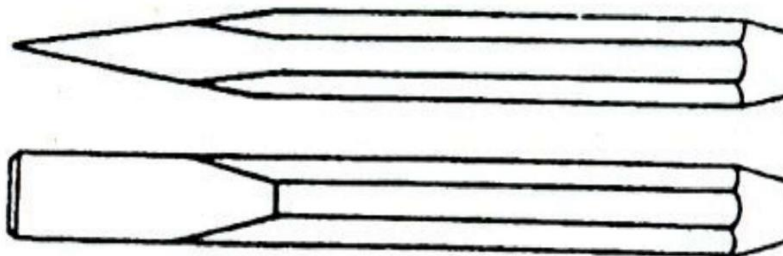


Figure: Flat chisel

Twist Drill

Twist drills are used for making holes. These are made of High-speed steel. Both straight and taper shank twist drills are used. The parallel shank twist drill can be held in an ordinary self-centering drill chuck. The taper shank twist drill fits into a corresponding tapered bore provided in the drilling machine spindle.

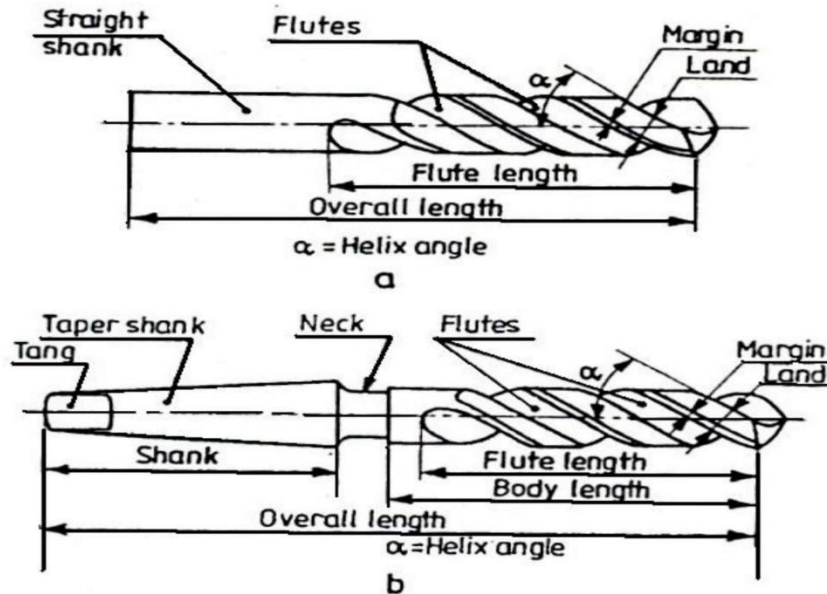


Figure: Twist drills

Taps and Tap wrenches

A tap is a hardened steel tool, used for cutting internal thread in a drill hole. Hand Taps are usually supplied in sets of three in each diameter and thread size. Each set consists of a taper tap, intermediate tap and plug or bottoming tap. Taps are made of high carbon steel or high-speed steel.

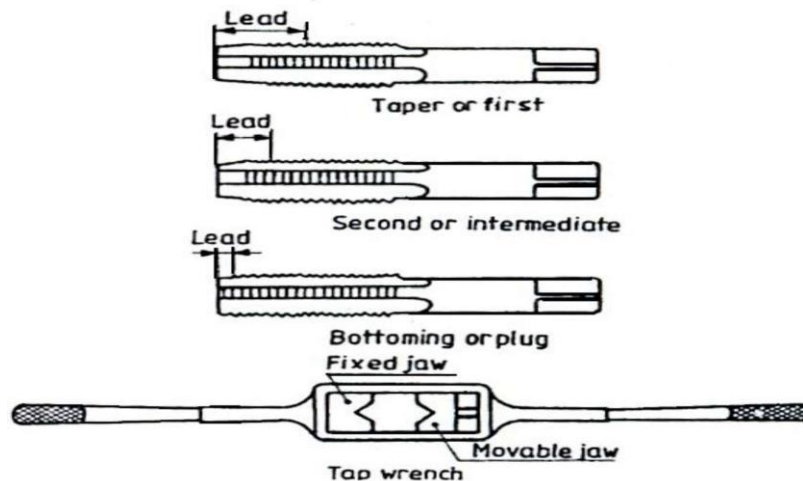


Figure: Taps and tap wrench

Dies and die-holders.

Dies are the cutting tools used for making external thread. Dies are made either solid or split type. They are fixed in a die stock for holding and adjusting the die gap. They are made of Steel or High Carbon Steel.

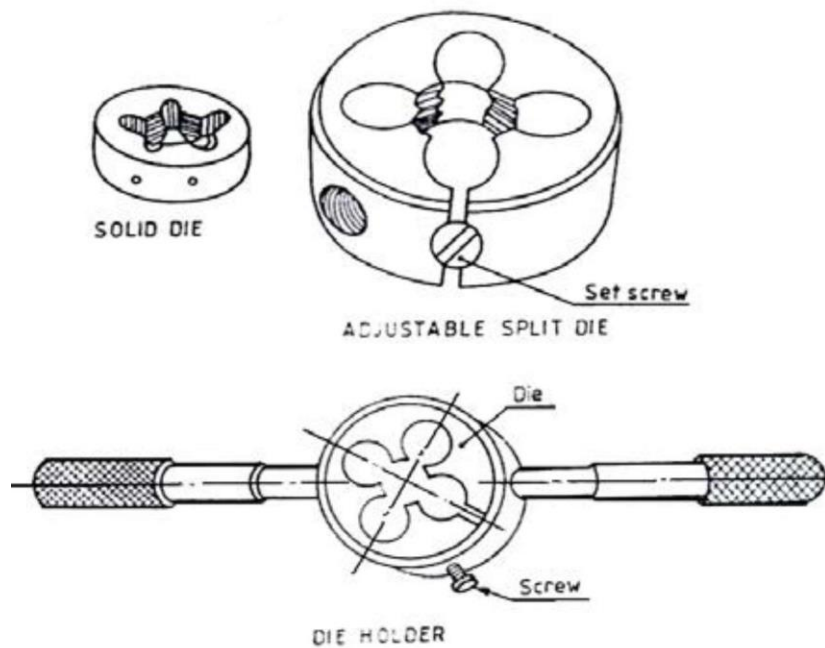


Figure: Dies and die holder

Bench Drilling Machine

Holes are drilled for fastening parts with rivets, bolts or for producing internal thread. Bench drilling machine is the most versatile machine used in a fitting shop for the purpose. Twist drills, made of tool steel or high-speed steel are used with the drilling machine for drilling holes.

Following are the stages in drilling work:

1. Select the correct size drills, put it into the chuck and lock it firmly.
2. Adjust the speed of the machine to suit the work by changing the belt on the pulleys. Use high speed for small drills and soft materials and low speed for large diameter drills and hard materials.
3. Layout of the location of the hole and mark it with a center punch.
4. Hold the work firmly in the vice on the machine table and clamp it directly on to the machine table.
5. Put on the power, locate the punch mark and apply slight pressure with the Feed Handle.

6. Once Drilling is commenced at the correct location, apply enough pressure and continue drilling. When drilling steel, apply cutting oil at the drilling point.
7. Release the pressure slightly when the drill point pierces the lower surface of the metal. This prevents the drill catching and damaging the work or drill.
8. On completion of drilling retrace the drill out of the work and put off the power supply.

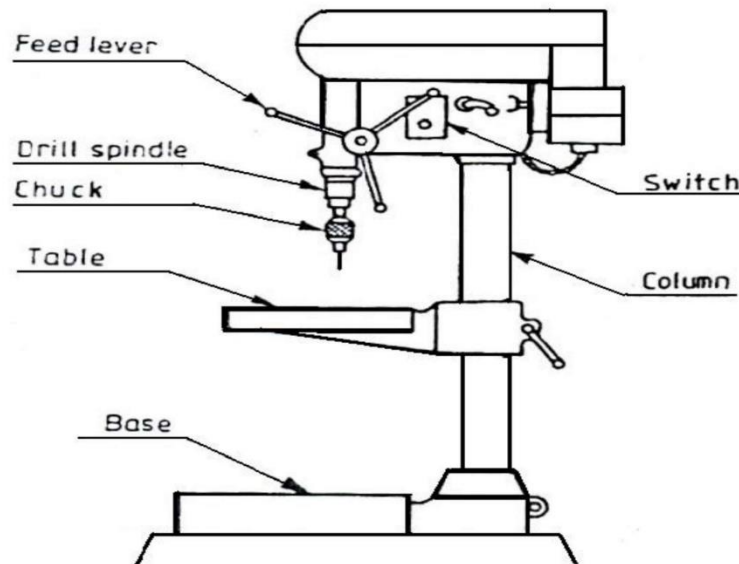


Figure: Bench drill

Finishing Tools

Reamers

Reaming is an operation of sizing and finishing a drilled hole, with the help of a cutting tool called reamer having several cutting edges. For this, a hole is first drilled, the size of which is slightly smaller than the finished size and then a hand reamer or machine reamer is used for finishing the hole to the correct size.

Hand Reamer is made of High Carbon Steel and has left hand spiral flutes so that, it is prevented from screwing into the whole during operation. The Shank end of the reamer is made straight so that it can be held in a tap wrench. It is operated by hand, with a tap wrench fitted on the square end of the reamer and with the work piece held in the vice. The body of the reamer is given a slight taper at its working end, for its easy entry into the whole during operation, it is rotated only in clockwise direction and also while removing it from the whole.

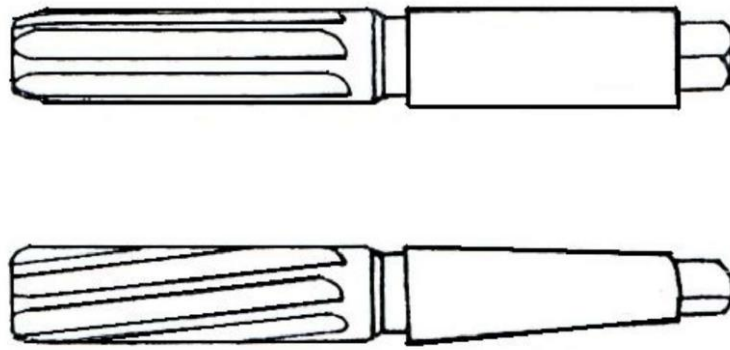


Figure: Reamers

Files

Filing is one of the methods of removing small amounts of material from the surface of a metal part. A file is hardened steel too, having small parallel rows of cutting edges or teeth on its surfaces. On the faces, the teeth are usually diagonal to the edge. One end of the file is shaped to fit into a wooden handle. The figure shows various parts of a hand file. The hand file is parallel in width and tapering slightly in thickness, towards the tip. It is provided with double cut teeth. On the faces, single cut on one edge and no teeth on the other edge, which is known as a safe edge.

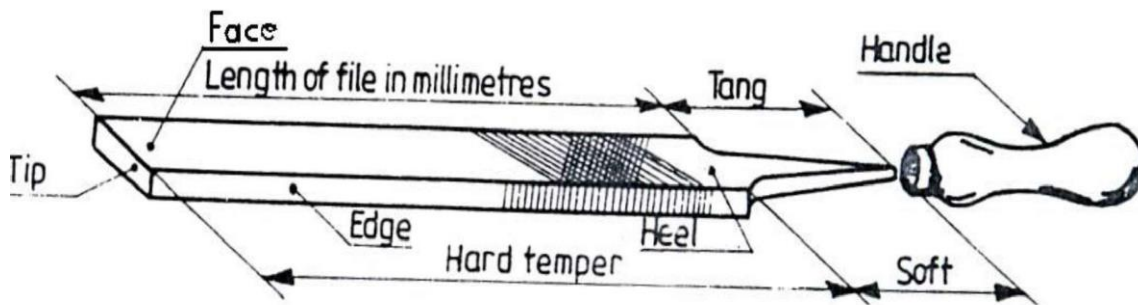


Figure: Parts of a hand file

Files are classified according to their shape, cutting teeth and pitch or grade of the teeth. The figure shows the various types of files based on their shape.

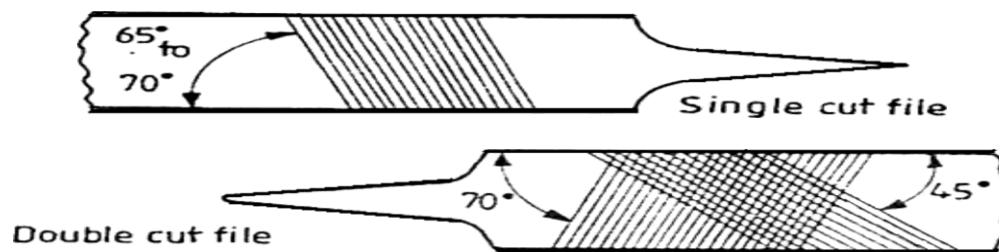


Figure: Single and double cut files

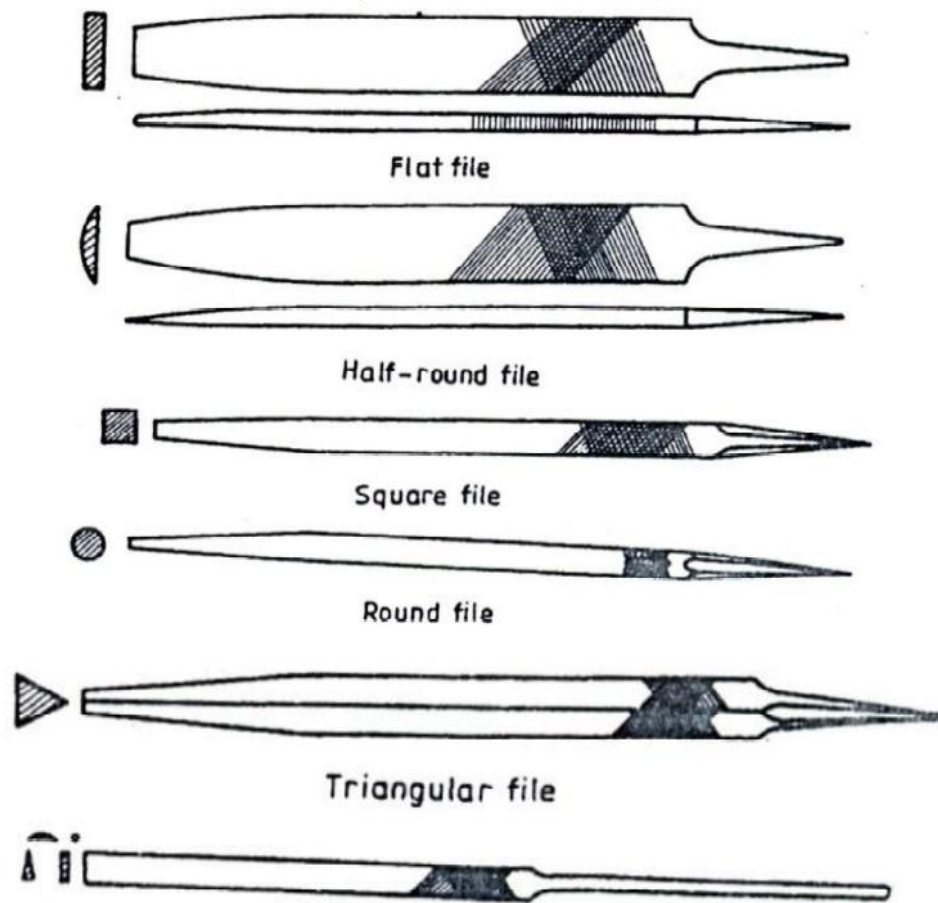


Figure: Types of files

Miscellaneous Tools

File card

It is a metal brush, used for cleaning the files, to free them from filings, clogged in between the teeth.

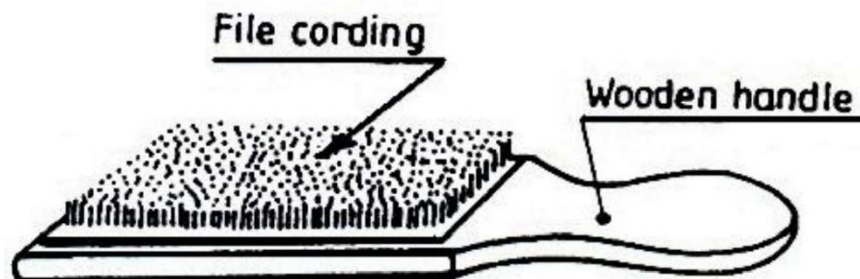


Figure: File card

Spirit level

It is used to check the leveling of machines.

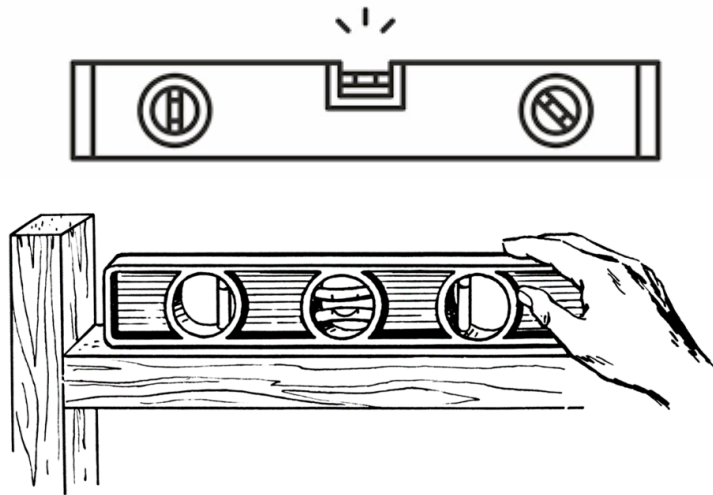


Figure: Spirit level

Ball Peen Hammer

Ball Peen Hammers are named, depending upon their shape and material and specified by their weight. A ball peen hammer has a flat face which is used for general work and a ball end, particularly used for riveting.

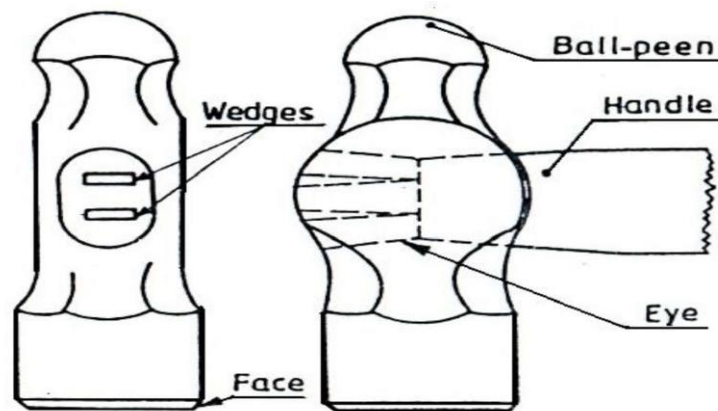


Figure: Ball peen hammer

Cross Peen Hammer

It is like ball peen hammer, except the shape of the peen. This is used for chipping, riveting, bending, and stretching metals and hammering inside the curves and shoulders.

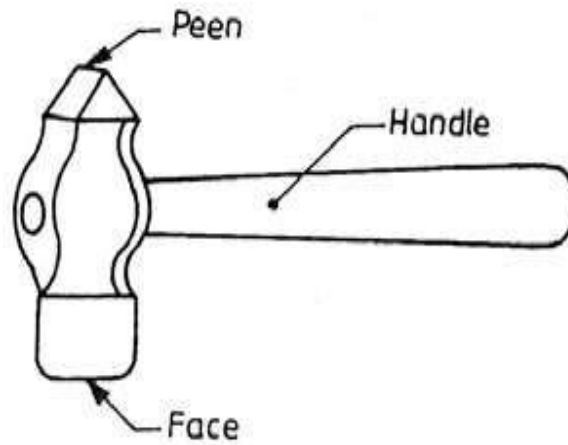


Figure: Cross Peen Hammer

Straight Peen Hammer

This is like cross peen hammer, but its peen is in line with the hammer handle. It is used for swaging, riveting in restricted places, and stretching metals.

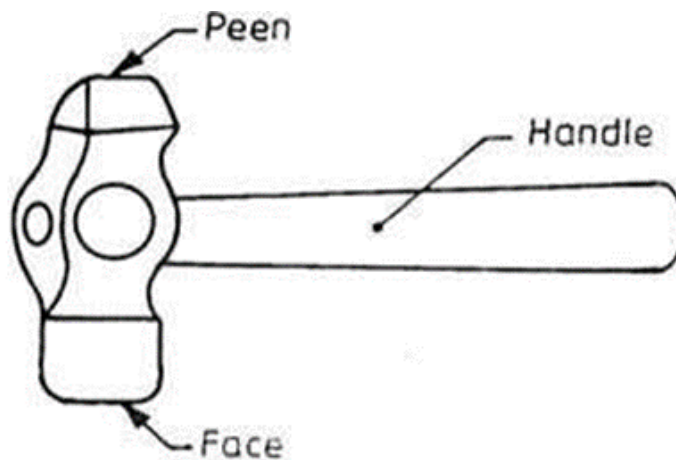


Figure: Straight Peen Hammer

Screwdriver

A screwdriver is designed to turn screws. The blade is made of steel and is available in different lengths and diameters. The grinding of the tip to the correct shape is very important. A star screwdriver is specially designed to fit the head of star screws. The end of the blade is fluted instead of flattened. The screwdriver is specified by the length of the metal part from handle to the tip.

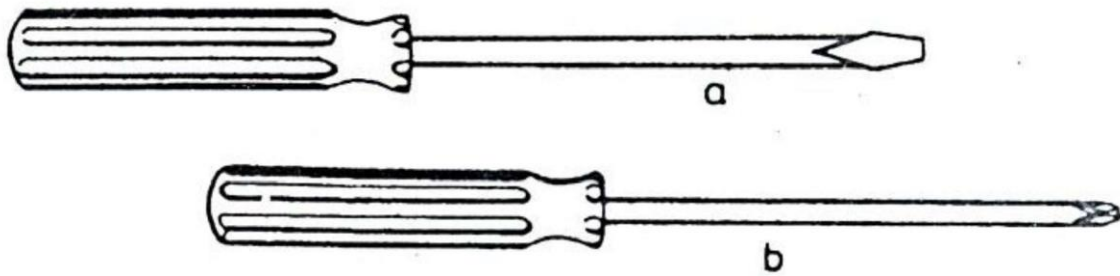
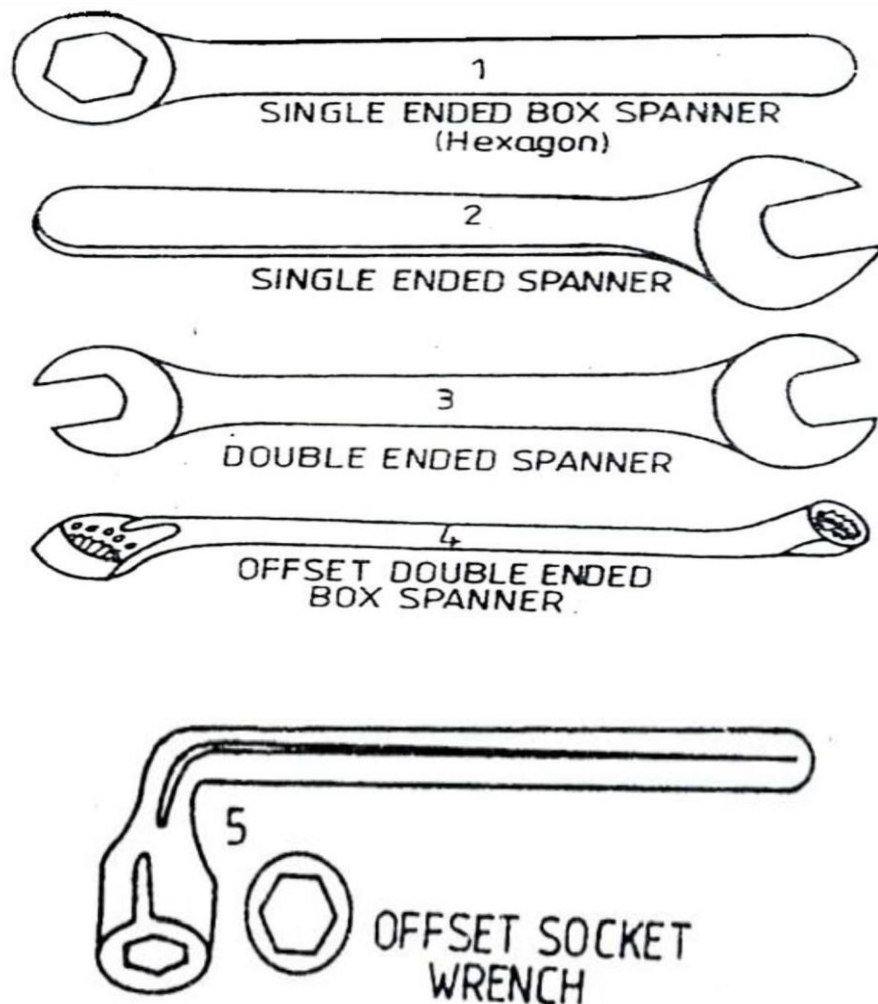


Figure: Screw drivers

Spanners

A spanner or wrench is a tool for turning nuts and bolts. It is usually made of forged steel. There are many kinds of spanners. They are named according to the application. The size of the spanner denotes the size of the bolt on which it can work.



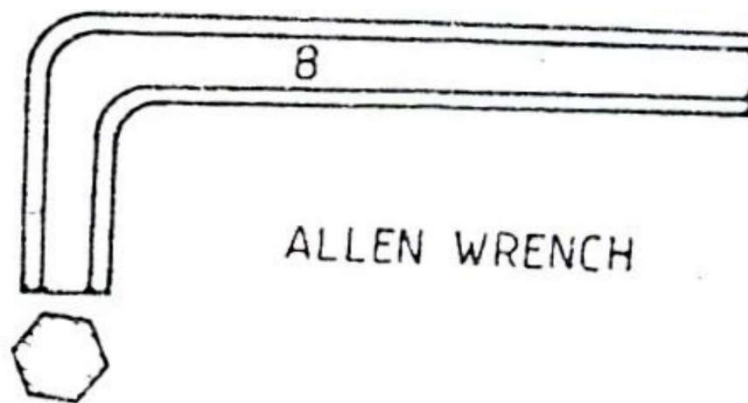
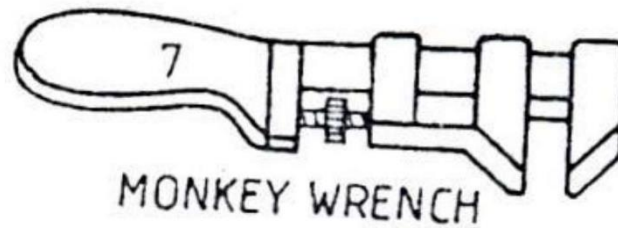
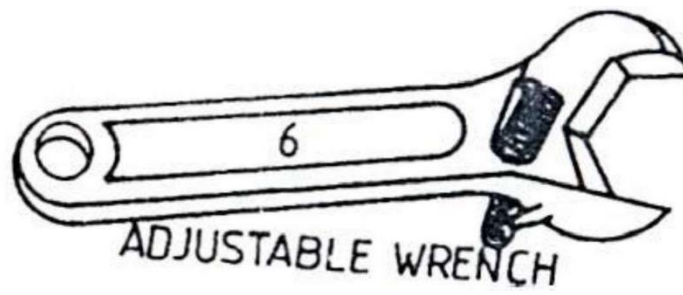
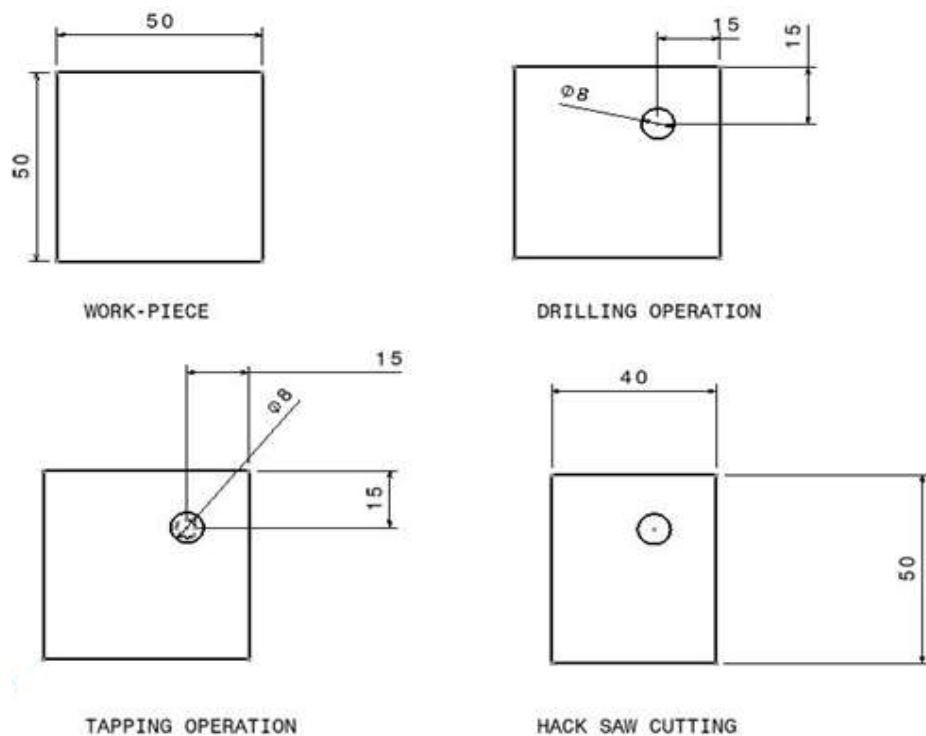


Figure: Spanners

Fitting Shop Job Description



Sequence of Operation

Marking- Marking means setting out dimensions as per the engineering drawing or directly transferring them from a similar part. Marking tools are used for this purpose.

Filing- Filing is a material removal process in manufacturing. Similar, depending on use, to both sawing and grinding in effect, it is functionally versatile, but used mostly for finishing operations, namely in deburring operations. Filing operations can be used on a wide range of materials as a finishing operation. Filing helps achieve work-piece function by removing some excess material and deburring the surface. Sandpaper may be used as a filing tool for other materials, such as glass.

The filing operation consists of the following steps:

1. The work should be held tightly in the vice. The amount of projection of the workpiece from the vice should be minimum to reduce the noise.
2. The file handle is held by the right hand. Left hand palm is pressed against the end of the file.
3. The file must remain horizontal throughout the stroke. The stroke should be long, slow, and steady. Pressure should be applied only in the forward direction.
4. The pressure is relieved during the return stroke, but the file should remain in contact with the workpiece.
5. When quantity of material removal is more, use rough files and for finishing cut use smooth files. Surface smoothness is generated progressively.
6. Filing is carried out to make the workpiece surface flat and to ensure perpendicularity

with the adjoining faces by removing a small amount of material. The perpendicularity between the adjoining faces is qualitatively checked by using a try square.

Drilling- The operation of making round holes in metal pieces is known as drilling. It is done with the help of a drilling machine.

The drilling operation is carried out as below:

1. The work is marked with a center punch at the center of the hole to be drilled.
2. Hold the job properly in a vice or in other suitable holding device.
3. The drill is fitted in the chuck.
4. The hand lever lowers the spindle with the chuck, and it is ensured that the point of the drill is in exact alignment with the previously marked center of the hole.
5. The pressure should be frequently relieved during the drilling operations, otherwise the cutting edges of the drill will be damaged.
6. A coolant should be used constantly during the drilling operations. Coolant is not used for drilling in Brass and Cast iron.
7. For drilling large size holes, first produce the small sized holes then produce large holes.

Tapping- The process of cutting internal threads into a drilled hole by using a tap is known as tapping. The procedure is described below:

1. A hole is drilled to a diameter smaller than the outside diameter of the thread on the tap.
2. After drilling, the taper tap is fixed in the tap wrench and screwed in the hole. The tap is held with its axis vertical.
3. The tap is not turned continuously, but after every half turn, it should be reversed slightly to clear the threads.
4. When the hole is through, the reduction of resistance on the tap indicates the cutting of a full thread by taper tap.
5. When many holes are to be tapped, in that case tapping is done by drilling machine with slow speed.

Sawing- It is the cutting operation to remove unwanted parts from the given workpiece. The procedure is as given below:

1. The work is held tightly in the vice.
2. The blade is fixed with its teeth facing forward so that the cutting stroke is forward stroke. The blade is tensioned sufficiently by a wing nut.
3. The desired cut is marked, and a notch is made with a file.
4. The frame is held by one hand just above the wing nut and the handle by another hand.
5. Cutting starts by keeping the blade slightly inclined.

6. Pressure is applied during the forward stroke only and it should be relieved during the return stroke.

Questions (Descriptive)

Q1 What is meant by fitting?

Ans Fitting is the process of assembling the item after bringing the components to the exact size and geometrical shape and tolerance

Q2 Which material is used to make the file?

Ans High carbon steel

Q3 What are the methods of filing and where it is used?

Ans Straight filing, cross filing and draw filing. straight filing is used to file thick materials and cross filing is used for filing thin material and to reduce the formation of convexity in the middle of the materials. draw filing is used for finishing purpose to remove any convexity. thickness

Q4 What are the types of vices? What is the name of vice used in fitting shop?

Ans Hand vice, pipe vice, bench vice, leg vice, Drill vice etc.

Bench vice

Q5 What is a try square?

Ans A try square is a wood-working or a metal working tool used for marking and measuring a piece of wood. The square refers to the tool's primary use of measuring the accuracy of a right angle (90 degrees); to try a surface is to check its straightness or correspondence to an adjoining surface. A piece of wood that is rectangular, flat, and has all edges (faces, sides, and ends) 90 degrees is called four squares.

Q6. Name of the operation used to cut thread inside a hole

Ans Tapping

Q7. What is the name of job material used in fitting shop?

Ans Mild steel

Q8. How to specify a file?

Ans By the shape, type, grade, and length.

Q9. List five types of tools used in Fitting shop.

Ans Bench vice

Hand Hacksaw

Hand File

Try square.

Tapping tool

Q10. How are files classified based on cut of teeth?

Ans The most used files according to cuts of teeth are:

Single cut

Double cut and

Rasp cut

Q11. What is the use of triangular files? What is the use for rasp files?

Ans Triangular files are commonly used for filing corners between 60° and 90°. They are double cut on all faces.

Rasp files are used for wooden sample and soft metals.

Q12. What is the material of scriber?

Ans High carbon steel

Q13. What is the use of Try Square?

Ans Try Square is used mainly for:

To check the flatness of a surface

To check the perpendicularity between two adjacent surfaces.

Q14. What is the use of square files?

Ans Square files are commonly used for filing in corners in jobs. They are double cut on all sides and tapers.

Q15 What is tapping? Name the tool used and the material of tool

Ans Operation for producing thread inside a hole. Taps made of high-speed steel (HSS) are used for the operation.

Q16. Distinguish between a single cut file and a double cut file.

Ans Single-cut files have rows of teeth cut parallel to each other at an angle of about 65 degrees from the centerline. Single-cut files are used for sharpening tools, finish filing and draw-filing. They are also the best tools for smoothing the edges of sheet metal.

Double-cut files have rows of teeth crisscrossed so teeth become diamond-shaped for fast cutting. Use double-cut files for quick removal of material and for rough work.

Q17. What are the different types of vices?

Ans Bench vice, Pin vice, leg Vice, Machine Vice, Pipe Vice, Angle vice, Rotary vice Saw Vice etc.

Q18. Mention any 3 types of files and their uses.

Ans Knife files: Used for slotting or wedging operations.

Round files: Used for enlarging round holes or cutting a scalloped edge.

Half round ring files: Used for filing inside of rings.

Q19. What should be the orientation of the hack hacksaw blade teeth to facilitate cutting by forward stroke

Ans Forward inclined

Q20. Why the hand hacksaw blade comprises of zig zag teeth?

Ans The zig zag nature of teeth facilitates efficient material removal during the cutting operation by eliminating the possibility of blade breakage.

Q21. What is the name of bench vice jaw plate material used in fitting shop? What is the name and purpose of the pattern made on the jaw plates?

Ans Mild steel

Knurled pattern for proper grip of the sample

Questions (Objective)

SN	Questions	A	B	C	D
1	Purpose for which Pin vice is used	Plumbing	Watch making	Key making	None
2	What is the purpose of vice	Holding Device	Cutting Device	Measuring Device	None
3	Files are classified based on	Shape	Grade	Cut	All of these
4	Rasp file can be used on material like	Lead	Wood	Brass	All of these
5	Type of file used for filing V groove	Triangular File	Half round file	Square file	Round file
6	The length of hacksaw blade is measured from	one end of the toothed portion to the other end	center of one pin hole to the center of another pin hole	end to end of blade	end of one hole to the end of the other pin hole
7	A rough file is used	on irregular jobs	to reduce the metal rapidly	on wood	if smooth file is not available
8	The tapping hole should be	larger than the tap size	smaller than the tap size	equal to the tap size	equal to the core (minor) diameter of the tap
9	Both sides of a flat bustard file have	no cut	single cut teeth	double cut teeth	wavy teeth
10	A file removes the metal during	forward stroke	both forward and return strokes	return stroke	none of these
11	Bench Vice is a tool used for	Cutting	Drilling	Holding	Boring
12	Tap are used for	Cutting accurate square holes	Finishing the round holes	Cutting internals threads in jobs	Cutting externals threads in jobs
13	For riveting purpose which hammer is used	Ball peen hammer	Cross peen hammer	Straight peen hammer	Double face hammer
14	What is the most common material for	Hard Steel	Hard grade Iron	Mild steel	High Carbon steel

	hacksaw blade?				
15	Which of these files has most no. of teeth per sq. inch?	Rough	Dead Smooth	Bastard	Smooth
16	Which tool is used for locating center for indentation mark for drilling purposes.	Try Square	Tap	Centre Punch	Hammer
17	All faces of triangular files are?	Single Cut	Double Cut	Rasp Cut	None
18	Tapping creates?	Holes	Internal threads	External threads	Flat Faces
19	Which tool will you use to check whether a square hole is truly square or not?	Scale	Mortise gauge	Try Square	File
20	Select the correct option representing parts of Try Square.	Blade and Stock	Blade and handle	Stock and handle	None of these
21	What is the least count of steel rule used in fitting shop?	0.50 mm	0.25 mm	1.0 mm	0.05 mm

Chapter 4

Computerized Numerical Control Shop

CNC is an acronym for Computer Numerical Control. A dedicated computer is used to perform all the basic NC functions. The complete part program to produce a component is input and stored in the computer memory and information for each operation is fed to the machine tools. The program can be stored and used in future.

Axis in CNC machines

The basis of axis identification is the 3-dimensional cartesian co-ordinate system, and three axes of movement are identified as x, y and z axis.

Z AXIS

The Z Axis of motion is always the axis of the main spindle of the machine. It does not matter whether the spindle carries the work piece or the cutting tool. On vertical machining centers Z axis is vertical and on horizontal machining center and turning centers Z axis is horizontal.

X-Axis

The axis is always horizontal and is always parallel to the work holding surface. Positive X Axis movement is identified as being to the right, when looking from the spindle towards its supporting column.

Y- Axis

The axis is always at a right angle to both X-Axis and Z-Axis.

CO-ORDINATE SYSTEM

Dimensional information in a work piece drawing can be stated in two ways Absolute Dimension System and Incremental Dimension System.

Absolute Dimension System

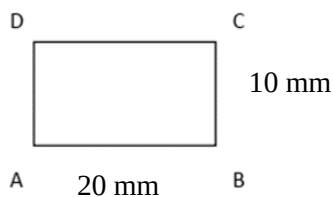
Data in absolute dimension system always refer to a fixed reference point. This point has the function of a coordinate zero point. The dimension lines run parallel to the coordinate axes and always start at the reference point. Absolute dimensions are also called as 'Reference dimensions.' In this absolute system, the set point is considered as a reference point as from that point, all the values are calculated, for example.

Point A = (0, 0)

Point B = (20, 0)

Point C = (20, 10)

Point D = (0, 10)



Incremental Dimension System

When using the Incremental Dimension system, every measurement refers to a previously dimensioned position. Incremental dimensions are the distance between adjacent points. These distances are converted into incremental coordinates by accepting the last dimension point as the coordinate origin for the new point. Incremental dimensions are also frequently called 'Relative dimensions' or 'Chain dimensions'. In this method, every point is considered as origin from this point; the values are calculated, for example.

Point A = (0, 0)

Point B = (20, 0)

Point C = (0, 10)

Point D = (-20, 0)

Programming methods:

In CNC machines program are programmed by two methods.

Manual part programming

Computer assisted part programming.

1) Manual part programming:

To prepare a part program using the manual method, the programmer writes the machining instruction must be hence, menu script the instruction must be prepared in a very precise manner because the typist prepares the NC type directory from the manuscript some in various form expending on the machine tool and tape format used.

2) Computer assisted part programming:

In the more complicated point and in contour application using manual part programming because an extremely tedious basic and subject to errors. It is must more appropriate to employ the high-speed digital computer to assist the part programming languages system that has been developed to automatically perform most of the calculation which the programmer would otherwise be forced to do.

CNC program structure

G-Codes: Preparatory Codes.

A number following address G determines the meaning of the command for the concerned block. G codes are divided into the following two types

Type	Meaning
One-shot G code	The G code is effective only in the block in which it is specified
Modal G code	The G code is effective until another G code of the same group is specified

G00: Rapid Travel

G01: Linear interpolation

G02: Circular interpolation CW

G03: Circular interpolation CCW

G04: Dwell for a specific time

G17: XY plane

G18: XZ Plane

G19: YZ plane

G20: Inch data input

G21: mm data input

G28: Go Home

G40: Cancel cutter Offset.

G41: Offset cutter Left.

G42: Offset cutter Right

G54: Workpiece coordinate system

G90: Program in absolute Coordinate

G91: Program in incremental Coordinate

I, J, K: Coordinate values of arc center corresponding to X, Y and Z axes.

F: Feed rate per minute or per revolution in either inches or mm

S: Spindle speed in RPM

T: Tool selection (used for machine with automatic tool changers)

R: Radius of arc which is used in circular interpolation

M-Code: Miscellaneous Code

These signals are used for ON/OFF control of a machine function such as tool change, spindle rotation change, coolant ON/OFF. One M code can be specified in one block. Selection of M codes for functions varies with the machine tool builder.

M01: Optional stop

M03: Spindle start CW

M04: Spindle start ACW

M05: Spindle Stop

M06: Tool Change

M07: Coolant on (Flood)

M08: Coolant on (Mist)

M09: Coolant off

M30: End of program

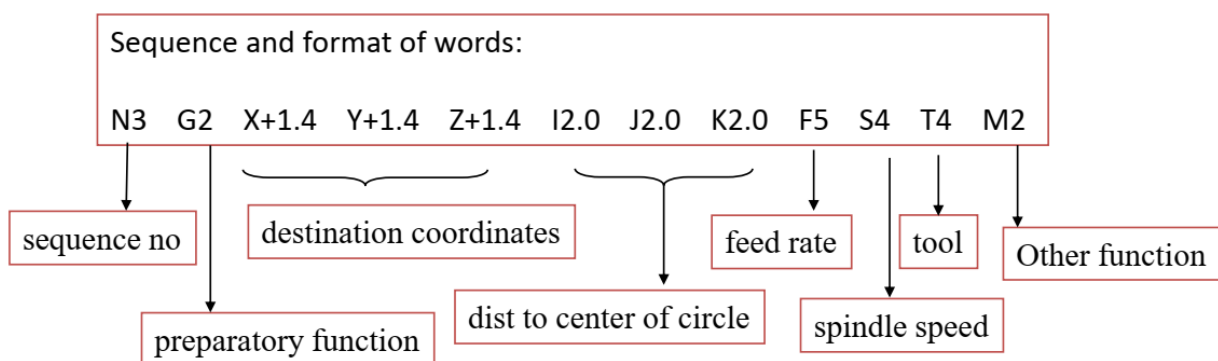


Figure: Sequence of part program

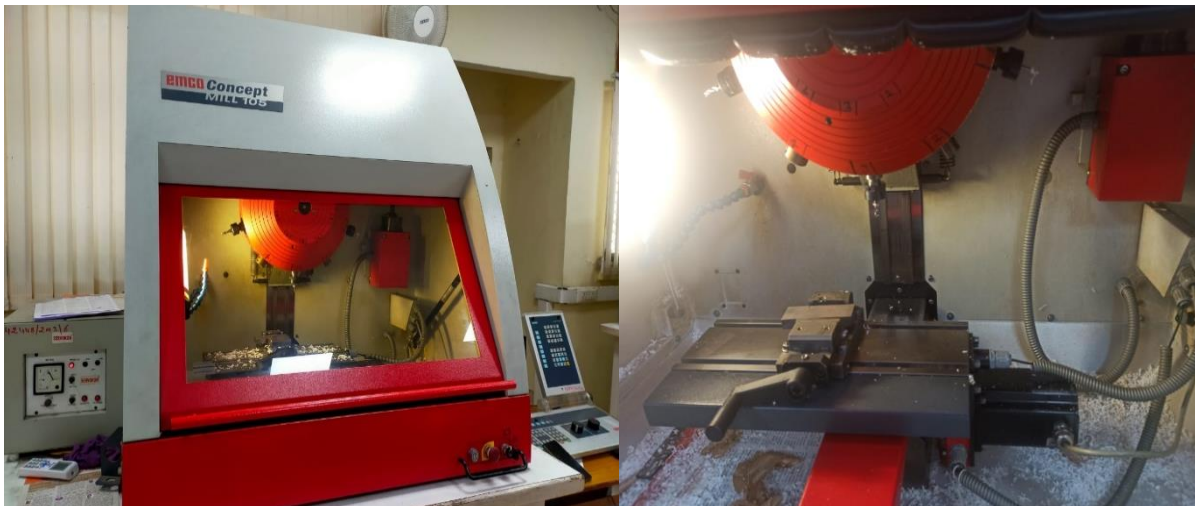


Figure: CNC Milling machine

Machine specifications:

WORK AREA

Travel in X/Y/Z	200 / 150 / 250 mm
Distance spindle nose	95 - 245 mm
Rapid motion speeds in X/Y/Z	5 m/min
Work feed X/Y/Z	0 - 5 m/min
Feed force in X/Y/Z	2000 N
Clamping area	420 x 125 mm
Max. table load	10 kg

MAIN SPINDLE

Max. speed	5000 rpm
Max. drive power	1.1 kW
Max. torque	4.2 Nm

Procedure for referencing

1. Start the spindle
2. Manually move the tool to a safe distance from the workpiece
3. Gradually adjust its position near to that face of the workpiece which is facing you.
4. Now turn on the jog mode and set the travel speed to 0.1mm/min.
5. Now slowly touch that face by jogging the tool in Y direction.
6. Upon touchup, Set Y zero at minus tool radius distance.
7. In the same method set X direction to zero.
8. For setting Z direction to zero, touch the upper surface of the workpiece by the tool in jogging mode and set to zero.

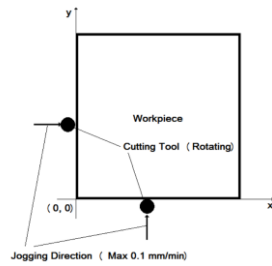
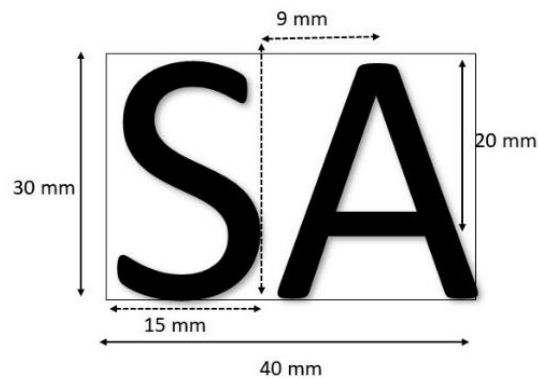


Figure 3: Workpiece zero-point referencing

CNC Job description:



G54	Workpiece coordinate system
T3 D1 M06	Tool 3 ready and tool change
S2000 M03	Spindle start in clockwise direction with 2000 RPM
G00 X0 Y0 Z5	Rapid movement to X0 Y0
G01 Z-8 F200	Linear interpolation with feed 200 mm/min
G00 Z3	to Z3
G00 X15 Y-7.5	Rapid movement, go to given coordinates
G01 Z-3	Linear interpolation, go to given coordinates
G03 X0 Y-7.5 CR7.5	Circular interpolation CCW ending at X0 Y-7.5 with radius 7.5 mm
G03 X7.5 Y-15 CR7.5	Circular interpolation CCW ending at X7.5 Y-15 with radius 7.5 mm
G02 X7.5 Y-30 CR7.5	Circular interpolation CW ending at X7.5 Y-30 with radius 7.5 mm

G02 X0 Y-22.5 CR7.5	Circular interpolation CW ending at X0 Y-22.5 with radius 7.5 mm
G00 Z3	Rapid movement to Z3
G00 X22 Y-30	Rapid movement, go to given coordinates
G01 Z-3	Linear interpolation, go to given coordinates
G00 X31 Y0	Rapid movement, go to given coordinates
G01 X40 Y-30	Linear interpolation, go to given coordinates
G00 Z3	Rapid movement, go to given coordinates
G00 X25 Y-20	Rapid movement, go to given coordinates
G01 Z-3	Linear interpolation, go to given coordinates
G01 X37 Y-2	Linear interpolation, go to given coordinates
G00 Z30	Rapid movement, go to given coordinates
M30	End of program

Task:

1. Write the part program for your name initials with comments on sheet and validate it with NC Viewer software.
2. Write the developed NC program in the machine console and make the key ring with your initials on an acrylic workpiece.

Questions

Objective

1. On a 3-axis CNC milling machine, rotational degree of freedom to the workpiece can be added as the 4th axis using an attachment. Say True or False

Ans: True

2. Which G code is preferred when no material removal is involved.

A. G00

B. G01

C. G02

D. G03

Ans: Option (A)

3. M codes are used for circular interpolations. Say True or False

Ans: False

4. G54 is used to set.

a. Tool Offset

b. Workpiece offset origin.

c. Tool radius offset.

d. Tool length compensation

Ans: Option (b)

5. M06 requires which compulsory command in the same line.

a. Tool magazine number T__

b. Spindle Speed S___

c. Feed F__

d. Depth of cut

Ans: Option (a)

6. Tolerances of the order $\pm 0.010\text{mm}$ can be achieved in CNC machining centers. Say True or False

Ans: True

7. Through hole of 50mm diameter in a direction perpendicular to the longitudinal axis of cylindrical workpiece of 100mm can be achieved using

a. Milling with linear interpolation

b. Milling with circular interpolation

c. Milling with helical interpolation

d. Boring with linear interpolation

Ans: Option (c)

8. Servo motors are used to control the table movement in a vertical machining center. Say True or False

Ans: True

9. Internal thread milling can be achieved by helical interpolation on vertical machining center. Say True or False.

Ans: True

10. What is the M-code for coolant ON/OFF?

a. M06/05

b. M07/08

c. M08/09

d. M10/M11

Ans: Option (c)

11. Which of the following is not the advantage of CNC machines?

a. Reduced scrap rate

b. Improved strength of the components

c. Higher flexibility

d. Improved quality

Ans: b

12. In CNC machine tool, the part program entered the computer memory.

a. can be used only once.

b. can be used again but it must be modified every time.

c. cannot say.

d. can be used again and again.

Ans: d

13. Which code is used for 'Circular interpolation clockwise direction'?

a. G01

- b. G02
- c. G03
- d. G00

Ans: b

14. Feed is measured in units of _____

- a. length/revolution
- b. degree/revolution
- c. length
- d. velocity

Ans: a

15. CNC machining centers do not include operations like _____

- a. milling
- b. boring
- c. welding
- d. tapping

Ans: c

16. In part programming, interpolation is used for obtaining _____ trajectory.

- a. helicoidal
- b. pentagonal
- c. triangular
- d. zigzag

Ans: a

17. The depth that the tool is plunged into the surface is called as _____

- a. feed
- b. depth of cut
- c. depth of tool
- d. working depth

Ans: b

18. In CNC systems multiple microprocessors and programmable logic controllers work _____

- a. in parallel
- b. in series
- c. one after the other
- d. for 80% of the total machining time

Ans: a

19. Point-to-point systems are used for _____

- a. reaming
- b. parting
- c. grooving
- d. facing

Ans: a

20. CNC machine has the following main advantage over conventional machining practice.

- a. Ability to employ higher cutting speeds, feeds, and depth of cut.
- b. Feedback control
- c. Flexibility
- d. None of the mentioned

Ans: b

Subjective

1. Write down the major components of a 3-axis vertical milling machine.

Ans: Table, spindle, work-holding fixture (vice), tool holding adapter, coolant pipe, automatic tool changer.

2. What is the use of an Automatic Tool Changer (ATC)?

Ans: The ATC helps to hold different types of tools at dedicated stations as defined by the user. A job may require 2 or more operations to be performed, therefore by using the M06 with T__ the respective tool from the tool station in the ATC can be called and the desired operation can be performed. Therefore, it eases operator effort and saves time involved in manual changing and offsetting of tools.

3. Mention any of the three most common types of work-holding devices that are used on a 3-axis

VMC?

Ans:

- a. Universal Vice
- b. Angle plate
- c. V-block

4. What are the two compensation techniques used on a 3-axis VMC?

Ans: Tool length compensation- G43 H__ (H01- 01 corresponds to the tool number at the ATC)

Tool radius compensation- G41 and G42

5. What are synchronous and asynchronous feeds?

Ans: In synchronous feed, the table/spindle movement depends on the RPM of the spindle and is denoted in mm/rev. In asynchronous feed rate, the table/tool movement is independent of the RPM of the tool and is denoted in mm/min.

6. How many types of CNC machines are there?

Ans: There are basically five different types of CNC machines:

- a. Plasma Cutting Machine.
- b. Laser Cutting Machine.
- c. Milling Machine.
- d. Router Machine
- e. CNC Lathe Machine

7. What language do CNC machines use?

Ans: The industry has standardized G-Code as its basic set of CNC machine codes. G-Code is the most popular programming language used for programming CNC machinery.

8. What is the difference between CNC milling and CNC lathe?

Ans: A lathe turns the material you're machining, and the tools are held stationary in the turret. Lathes are used mainly for round parts. Milling machines hold the material stationary and the tools do the movement. Milling machines are used for any shape part.

9. What Is Meant by Canned Cycle?

Ans: A Canned cycle is a way of conveniently performing repetitive CNC machine operations. Canned cycles automate certain machining functions such as drilling, boring, tapping, pocketing, etc... Canned cycles are so called because they allow a concise way to program a machine to produce a feature of a part.

10. What Is the Machine Control Unit?

Ans: The machine control unit (MCU) is the heart of a CNC system. It is used to perform the following functions: To read the coded instructions. To decode the coded instructions. To implement interpolations (linear, circular, and helical) to generate axis motion commands.

Chapter 5

Welding Shop

Definition of Welding:

Welding is the process of joining similar metals by the application of heat, with or without application of pressure or filler metal, in such a way that the joint is equivalent in composition and characteristics of the metals joined. In the beginning, welding was mainly used for repairing all kinds of worn or damaged parts. Now, it is extensively used in manufacturing industry, construction industry (construction of ships, tanks, locomotives, and automobiles) and maintenance work, replacing riveting and bolting, to a greater extent.

In short, the welding can be done:

- with the help of filler material or without filler material.
- with the help of pressure and non-pressure.

Homogeneous Welding – Similar Nature of Material Joining

Heterogeneous Welding- Dissimilar Nature of Material Joining

Welding Societies:

AWS: American Welding Society

IIW: Indian Institute of Welding

IWS: Indian Welding Society

Importance of Welding:

1. No automobile industry can be assumed without welding.
2. Satellite, Aeronautical Industries (Majorly requires Dissimilar joining)
3. Ship Building Industries, Gas Pipeline, Oil Pipeline Etc.

Classification of welding process:

Welding processes are of several types, these can be broadly classified into following categories:

1. Electric Arc Welding
2. Gas Welding
3. High Energy Beam Welding
4. Resistance welding
5. Thermite Welding (Railway Track are joined chemically)
6. Solid state welding

Further classification the above-mentioned welding processes is given below:

1. Electric Arc Welding:
2. SMAW –Shielded Metal Arc Welding/ MMAW-Manual Metal Arc Welding/ Stick Welding
3. MIG/GMAW-Metal Inert Gas Welding/Gas Metal Arc Welding
4. TIG/ GTAW -Tungsten Inert Gas Welding/Gas Tungsten Arc Welding
5. Gas Welding:
6. OAW-Oxy Acetylene Welding
7. OHW-Oxy Hydrogen Welding
8. High energy beam welding:
9. Laser welding
10. Electron beam welding
11. Resistance Welding:

12. Spot Welding
13. Seam Welding (Continuous)
14. Butt Welding
15. Projection Welding
16. Solid state welding:
17. Friction welding
18. Ultrasonic welding
19. Soldering and brazing

Possible hazards in the Welding Shop:

Welding “smoke” is a mixture of very fine particles (fumes) and gases. Many of the substances in welding smoke, such as chromium, nickel, arsenic, asbestos, manganese, silica, beryllium, cadmium, nitrogen oxides, phosgene, acrolein, fluorine compounds, carbon monoxide, cobalt, copper, lead, ozone, selenium, and zinc, can be extremely toxic. Generally, welding fumes and gases come from: the base material being welded or the filler material that is used. Coatings and paints on the metal being welded, or coatings covering the electrode. Shielding gases supplied from cylinders. Chemical reactions which result by the action of ultraviolet light from the arc and heat. Process and consumables used, and contaminants in the air, for example vapors from cleaners and degreasers.

The health effects of welding exposures are difficult to list because the fumes may contain so many different substances that are known to be harmful. The individual components of welding smoke can affect any part of the body, including the lungs, heart, kidneys, and central nervous system. Welders who smoke may be at greater risk of health impairment than welders who do not smoke, although all welders are at risk.

Exposure to welding smoke may have short-term and long-term health effects. These effects are described below:

Short-Term (Acute) Health Effects:

Exposure to metal fumes (such as zinc, magnesium, copper, and copper oxide) can cause metal fume fever. Symptoms of metal fume fever may occur 4 to 12 hours after exposure, and include chills, thirst, fever, muscle ache, chest soreness, coughing, wheezing, fatigue, nausea, and a metallic taste in the mouth.

Welding smoke can also irritate the eyes, nose, chest, and respiratory tract, and cause coughing, wheezing, shortness of breath, bronchitis, pulmonary edema (fluid in the lungs) and pneumonitis (inflammation of the lungs). Gastrointestinal effects, such as nausea, loss of appetite, vomiting, cramps, and slow digestion, have also been associated with welding.

Some components of welding fumes, for example cadmium, can be fatal in a short time. Gases given off by the welding process can also be extremely dangerous. For example, ultraviolet radiation given off by welding reacts with oxygen and nitrogen in the air to form ozone and nitrogen oxides. These gases are deadly at high doses and can also cause irritation of the nose and throat and serious lung disease.

Ultraviolet rays given off by welding can react with chlorinated hydrocarbon solvents, such as 1-trichloroethane, trichloroethylene, methylene chloride, and perchloroethylene, to form phosgene gas. Even a very small amount of phosgene may be deadly, although early symptoms of exposure -- dizziness, chills, and cough -- usually take 5 or 6 hours to appear. Arc welding should never be performed within 200 feet of degreasing equipment or solvents.

Long-Term (Chronic) Health Effects:

Studies of welders, flame cutters, and burners have shown that welders have an increased risk of lung cancer, and possibly cancer of the larynx (voice box) and urinary tract.

Welders may also experience a variety of chronic respiratory (lung) problems, including bronchitis, asthma, pneumonia, emphysema, pneumoconiosis (refers to dust-related diseases), decreased lung capacity, silicosis (caused by silica exposure) and siderosis (a dust-related disease caused by iron oxide dust in the lungs).

Other health problems that appear to be related to welding include heart disease, skin diseases, hearing loss and chronic gastritis (inflammation of the stomach), gastroduodenoscopy (inflammation of the stomach and small intestine) and ulcers of the stomach and small intestine. Welders exposed to heavy metals such as chromium and nickel have also experienced kidney damage.

Welding also poses reproductive risks to welders. A recent study found that welders, and especially welders who worked with stainless steel, had poorer sperm quality than men in other types of work. Several studies have shown an increase in either miscarriages or delayed conception among welders or their spouses. Possible causes include exposure to: (1) metals, such as aluminum, chromium, nickel, cadmium, iron, manganese, and copper, (2) gases, such as nitrous gases and ozone, (3) heat and (4) ionizing radiation (used to check the welding seams). Welders who perform welding or cutting on surfaces covered with asbestos insulation are at risk of asbestosis, lung cancer, mesothelioma, and other asbestos-related diseases. Employees should be trained and provided with the proper equipment before welding near materials that contain asbestos.

Safety Gear & Purpose:

1. Welding Handheld Face Shield (UV Protection)
2. Strong Leather Shoes (Electric Safety)
3. Leather gloves (Protection from thermal/heat)
4. Metal Tong (Hot Metal/Object Handling)
5. Lab Coat (Hot spatters/Flying Metal Particles)
6. Safety Goggles (Used during Slag removal/ Chipping process)



Figure. Schematic of safety gear for a welding process

Description of welding processes being used for demonstration:

Resistance spot welding:

Basic introduction and principle:

Resistance Spot welding is comparatively modern welding process. It came into the field of welding in between the years 1900 - 1905. It is the most widely used resistance welding method. The main purpose of resistance spot welding method is to join two to four light gauge overlapping metal sheets (which can have thickness up to 3 mm).

At first the job is cleaned and all types of contaminants like grease, oil, dirt, scale, and paint are removed. The surface of the electrodes are also very clean. For clamping the metal sheets together two copper electrodes are used at the same time. The current passes through electrodes and then into the metal sheets. Because of the resistance, heat is generated in the air gap within the contact points. Since copper is a great conductor heat dissipates to the metal so quickly. As the metal (workpiece) is a poor conductor of heat in comparison to the copper electrode the heat remains in the air gap.

So, the heat remains in one place creating a strong effect and the metal is melted at that desired spot. The period of heat dissipation is very small and currently metal gets melted and then becomes solid and thus the joint is formed.

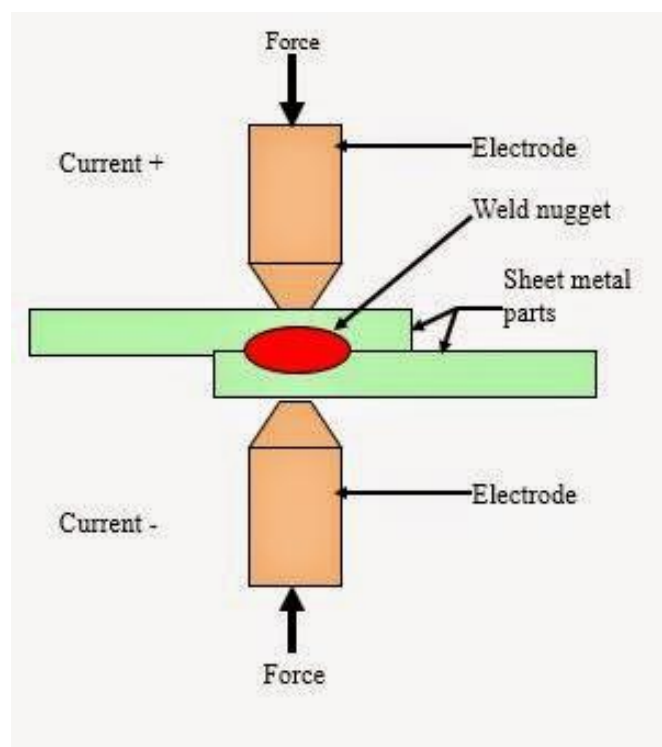


Figure. Schematic of resistance spot welding.

Steps involved in Resistance Spot Welding are shown in the following diagram.

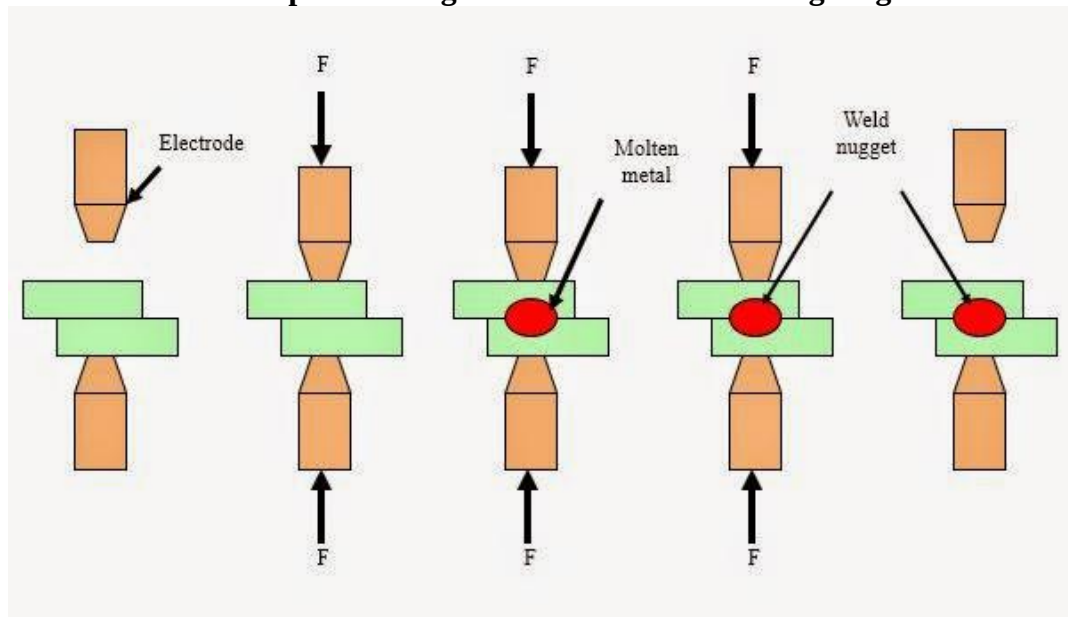


Figure. Various steps involved to form a joint using resistance spot welding

Advantages of Resistance Spot Welding:

Comparatively Low cost

Resistance Spot Welding (RSW) method doesn't need highly skilled workers.

Distortion or warping of parts is eliminated though it leaves some depressions or indentation.

The joint made is highly uniform.

Automatic or semi-automatic operation both can be done.

There is no need for edge preparation.

Welding can be done in quick succession. It just needs a few seconds to make the joint.

Disadvantages of RSW:

The equipment cost is high so it can influence the initial cost.

Skilled welders or technicians are needed for the maintenance and controlling.

Some metals need special surface preparation for making the RSW a success.

The thick jobs are not easy to weld.

Applications of Resistance Spot Welding:

Spot welding of thick steel plates has been done and it has replaced the need for riveting.

The welding of two or more sheet metals can be joined by mechanical means more economically by using the spot-welding methods. We don't need gas tight joints.

Spot welding can be used for attaching braces, pads, or clips with cases, bases and covers which are mainly products of sheet metal forming.

Automobile and aircraft industries rely greatly of spot welding these days.

Electric arc welding/ Shielded metal arc welding (SMAW):

Basic introduction:

Arc welding or shielded metal arc welding (SMAW) is the welding process, in which heat is generated by an electric arc struck between a consumable electrode and the work piece. Electric arc is luminous electrical discharge between two electrodes through ionized gas. Any arc welding method is based on an electric circuit consisting of the following parts:

1. Power supply (AC or DC).
2. Welding electrode.

3. Work piece.

Welding leads (electric cables) connecting the electrode and work piece to the power supply.

Electric arc between the electrode and work piece closes the electric circuit. The arc temperature may reach 10000°F (5500°C), which is sufficient for fusion of the work piece edges and joining them. When a long joint is required, the arc is moved along the joint line. The front edge of the weld pool melts the welded surfaces when the rear edge of the weld pool solidifies forming the joint. Transformers, motor generators and rectifiers' sets are used as arc welding machines. These machines supply high electric currents at low voltage and an electrode is used to produce the necessary arc. The electrode serves as the filler rod and the arc melts the surface so that the metals to be joined are fixed together.

Arc Welding Equipment:

1. Electrode holder- Electrode holder is used for holding the electrode manually and conducting current to it. These are usually matched to the size of the lead, which in turn matched to the amperage output of the arc welder. Electrode holders are available in sizes that range from 150 to 500 Amps.
2. Welding Electrodes- An electrode is a piece of wire or a rod of a metal or alloy, with or without coatings. An arc is set up between electrode and workpiece.
3. Hand Screen- Hand screen used for protection of eyes and supervision of weld bead.
4. Chipping hammer- Chipping Hammer is used to remove the slag by striking.
5. Wire brush -Wire brush is used to clean the surface to be welded.

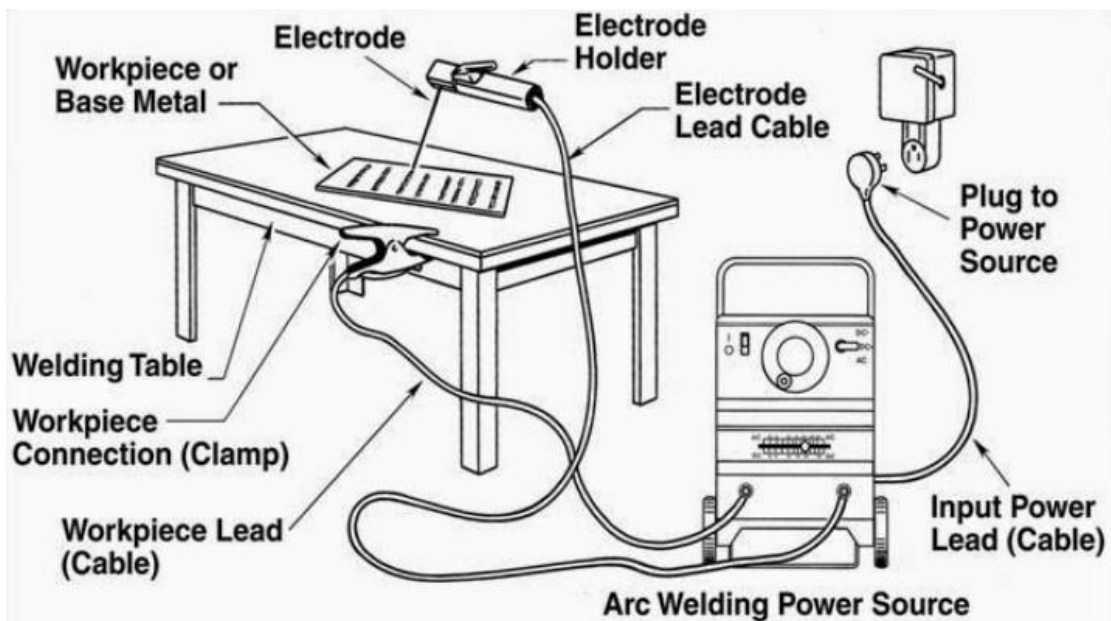


Figure. Schematic of a shielded metal arc welding setup

Safety Recommendations for ARC Welding:

Beginners in the field of arc welding must go through and become familiar with these general safety recommendations which are given below.

The body or the frame of the welding machine shall be efficiently earthed. Pipelines containing gases or inflammable liquids or conduits carrying electrical conductors shall not be used for a ground return circuit. All earth connections shall be mechanically strong and electrically adequate for the required current. Welding arc in addition to being very is a source of infra-red and ultra-violet light also; consequently, the operator must use either helmet or a hand-shield fitted with a special filter glass to protect eyes.

Excess ultra-violet light can cause an effect like sunburn on the skin of the welder.

Gloves protect the hands of a welder.

A leather or asbestos apron is very useful to protect welder's clothes and his trunk and thighs while seated he is doing welding.

Gas welding

Brief introduction:

Oxygen and acetylene together in a flame provide the heat necessary to melt most metals. This combined with a neutral welding atmosphere and suitable filler material allows a skilled operator to weld most metals. Other fuel gases such as LPG or propane produce a reactive secondary flame that interferes with the molten metal, making them unsuitable for welding. These and other fuel gases are suitable for heating and cutting purposes.

Applications:

1. It is used to join thin metal plates.
2. It can be used to join both ferrous and non-ferrous metals.
3. Gas welding is mostly used in fabrication of sheet metal.
4. It is widely used in automobile and aircraft industries.

Components of gas welding setup:

Main parts of a Oxyacetylene gas welding system is shown in figure below.

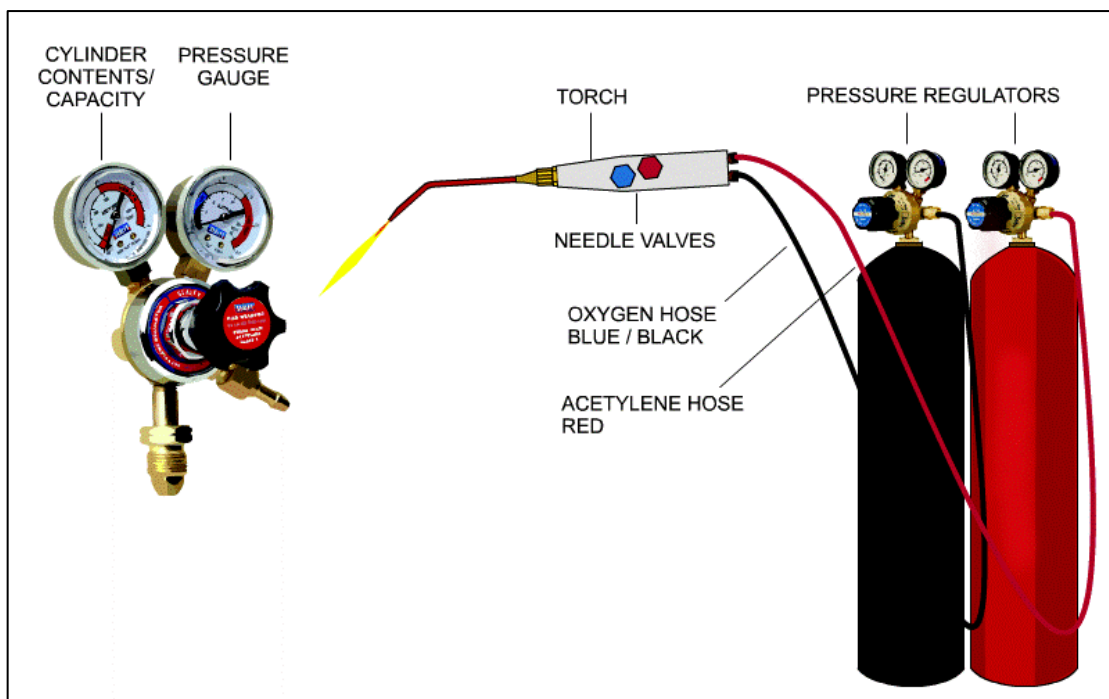


Figure. A gas welding (oxy-acetylene welding) setup

1. Torch- It mixes oxygen and acetylene to produce flame and gives direction to the flame.
2. Oxygen hose-It is a rubber tube used to carry oxygen from oxygen cylinder to the torch. Black or blue in color.
3. Acetylene hose-It transports Acetylene from cylinder to the torch. Red in color.
4. Pressure regulators-These are used to regulate flow(pressure) of Oxygen and Acetylene in the

respective hoses.

5. Cylinders-Cylinders carry gases at high pressure. Black color cylinder is used for Oxygen and maroon color is used for acetylene.

Safety precautions:

1. Always wear protective clothing, i.e., flame retardant overalls.
2. Always wear the correct eye goggles.
3. Always have the spindle key in the acetylene cylinder valve.
4. Always keep cylinders secured in an upright position.
5. Always check for leaks with a soapy solution, NEVER with a naked flame.
6. Never carry out makeshift repairs on welding equipment.
7. Never allow oil or grease to come in contact with oxygen equipment.
8. Never weld an enclosed vessel, i.e., petrol / oil drums until they have been thoroughly cleaned.
9. Never work in an enclosed vessel on your own and always leave the cylinders outside. If working in an enclosure vessel, adequate ventilation should be provided, and firefighting equipment should be available.
10. In the event of a serious flashback or backfire plunge the blowpipe in a bucket of cold water, leaving the oxygen running to prevent water entering the blowpipe.
11. Should the hoses become damaged, turn off the supply of gas at the cylinder.
12. Don't forget, this equipment, if misused or damaged, can be dangerous.
- 13.

Advantages:

1. It is easy to operate and does not require a skilled operator.
2. Equipment cost is low compared to other welding processes like MIG, TIG etc.
3. It can be used on site.
4. Equipment is more portable than other types of welding.
5. It can also be used as gas cutting.
6. Disadvantages:
7. It provides a low surface finish. This process needs a finishing operation after welding.
8. Gas welding has large heat affected zones which can cause change in mechanical properties of parent material.
9. Higher safety issues due to naked flame of high temperature.
10. It is Suitable only for soft and thin sheets.
11. Slow metal joining rate.
12. No shielding area which causes more welding defects

JOB DESCRIPTION:

Types of Joint:

Lap Joint

Butt Joint

T joint

Corner Joint

Edge Joint

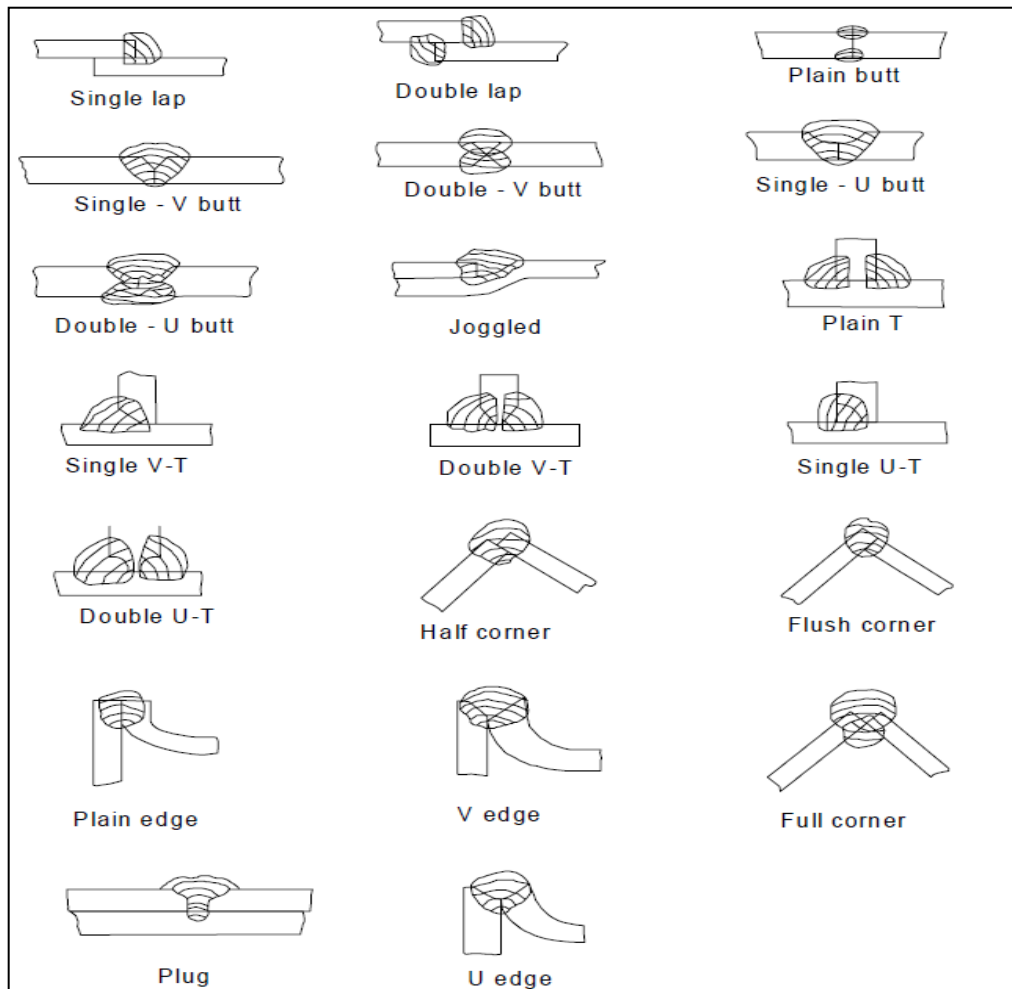


Figure. Various types of welding joints

Welding Position:

1. Flat Position
2. Horizontal Position
3. Vertical Position
4. Overhead Position



Figure. Welding positions

Resistance Spot Welding:

1. Work-piece Material: Galvanized Steel sheets (steel sheets which have been coated with zinc).
2. Workpiece Size: 80x30 x0.3 mm two sheets.
3. Electrode Used: Copper or Tungsten and Molybdenum.

SMAW:

Work-piece Material: Mild Steel/Low Carbon Steel (Carbon Content: 0.15-0.45 percent)

Work-piece Size: 50x50 x6 mm

Electrode Used (3.15x350mm 5 diameters (core metal)

Type of Flux on Electrode: Ferromanganese, rutile, cellulose, potassium, etc.

Specification of 6013 Electrode:

The initial two digits represent tensile strength (Needs to multiply by 1000): 60000 PSI-Pound per Square Inch.

The Third digit represents the welding position (Could be used in all positions)

The last digit denotes the type of flux coating. 03 stands for ferromanganese, cellulose, rutile, potassium silicate as a binder, and other siliceous materials.

An operation that can be performed: It can be used for welding ferrous material (including steel, stainless steel, and cast iron).

Formula to calculate current: 35D (D stands for electrode diameter in millimeters)

Base Metal Thickness	Electrode Size in mm	Welding Amperes
1.6-3.2 mm	2.5 mm	50-90 A
3.2-6.4 mm	3.15 mm	90-140 A
6.4-9.5 mm	4 mm	120-180 A
9.5-19 mm	5 mm	150-230 A
19-25.4 mm	6 mm	200-300 A

Welding Voltage:

OCV (Open Circuit Voltage)-80 Volt: It helps in easy ignition of the arc.

CCV (Close Circuit Voltage)-15-40 Volt: It is associated with Air Gap and increases with an air gap.

Current (Amp)	Voltage
Up-to 100	15
100-200	20
200-250	25
250-350	30
350-500	35
Over 500	40

Welding Speed: Based on the thickness of the base metal and Weld-Bead requirement.

General Instructions during working for the Welding Practical:

Do's:

1. In the Welding workshop, 04 welding platforms are available. Groups will be formed out of total lab class strength accordingly.
2. Before working, make sure no user has cellphone, ring, Analog/Digital wristwatch, etc. Please switch off your phones and smartwatches and then put them in your bag.
3. Two Students will share one single job. But please make a note that one side will be utilized by everyone for final job work.
4. Please make sure that your viva voice (Oral Examination) has been conducted by TA during the practical class hours. Please don't leave the class until it is over.
5. Please write entry no. on the work piece with the Marker. Please make sure it should be legible.
6. Please keep all the tooling and safety gear carefully at the designated place after the job submission. It's a collective responsibility.
7. Please remove the lab coat when leaving the lab and hang it properly at
8. designated place.
9. Please seek the Lab- instructor permission when leaving the class.

Don't s:

1. Never keep the plastic-made (handle) chipping hammer in the hot working zone.
2. Never throw leftover electrodes on the ground. It may burn user shoes in hot conditions.
3. Never use/keep any wet objects on the working table.
4. Never utilize the tong/s in welding tasks that are wet.
5. Never Keep the Face Shield on the Ground. Other students may trip over the shield and its glass may break.
6. Never keep the electrode holder on the working table.
7. Never use Torn gloves in welding activities.

8. Never use a Marker on the hot work- piece for entry no. recording.

JOB DESCRIPTION WITH DIAGRAMS

Bead on plate (BOP):

1. Workpiece dimensions for BEO:
Length: 100 mm
Width: 50 mm
Thickness: 6 mm
2. Workpiece material: Mild steel
3. Electrode type and diameter: E6013, 3.15 mm

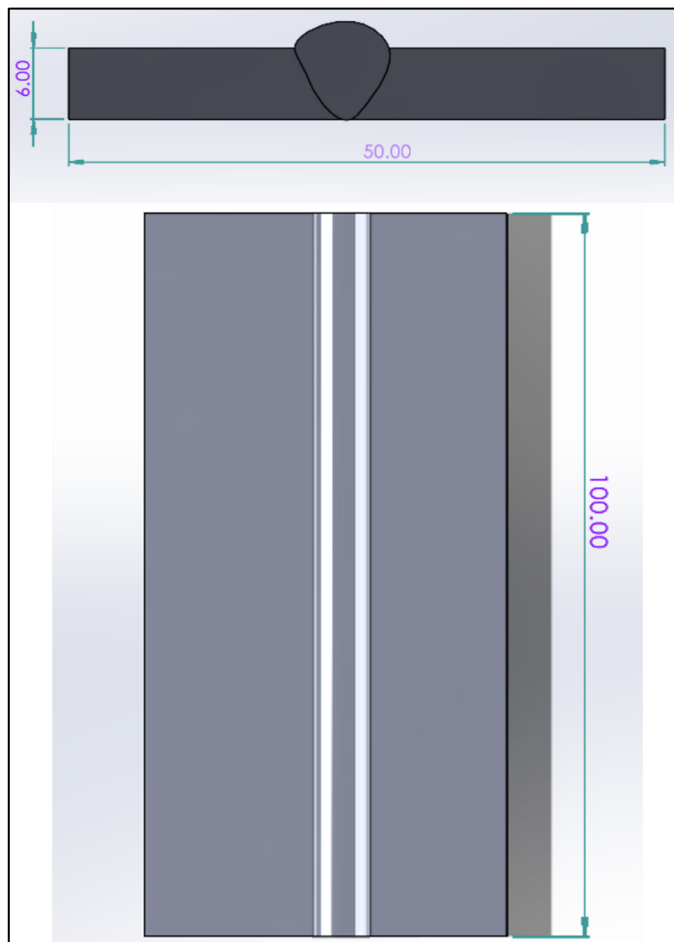


Figure. Workpiece schematic for bead on plate experiments

Butt joint:

- Workpieces: 2
Length of each plate: 100 mm
Width: 50 mm
Thickness: 6 mm
Workpiece material: Mild steel
Process: SMAW
Electrode type and diameter: E6013, 3.15 mm

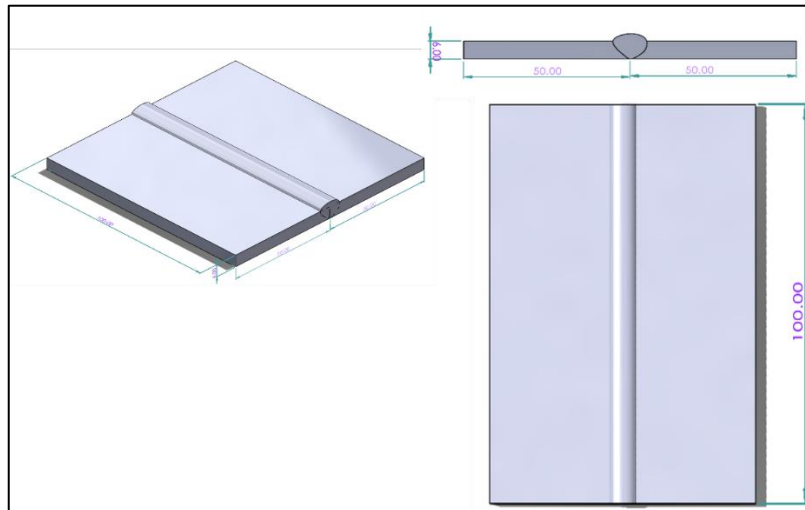


Figure. Workpiece schematic and dimensions for butt joint

General questions and answers related to welding:

1. What is welding?
Answer: it is a process in which heat and/or pressure can be applied together to join two sheets or plates.
2. In shielded metal arc welding (SMAW), which type of electrode is used?
Answer: flux coated consumable electrode
3. What is the use of flux coating on welding electrode?
Answer: flux coating is employed to protect the molten metal from atmospheric gases to avoid defects, it also helps in stabilizing the welding arc.
4. Describe the electrode specifications used in SMAW process?
Answer: for example, a typical SMAW electrode is designated as E6013
where, E (first letter): Extruded rod.
60(2nd and 3rd letters): Strength of electrode in Psi.
1(4th letter): Welding positions in which electrode can be used.
3 (5th or last letter): Type of coating.
5. What is the purpose of chipping hammer?
Answer: it is generally used to remove the slag deposited over the weld bead.
6. In arc welding which type of radiation is produced that affects the human eyes severely?
Answer: ultraviolet radiations
7. What is an arc eye?
Answer: arc eye is the physical condition in which welder's eyes are red and having severe needle like pain and irritation in eyes. It happens due to the exposure of eyes to radiation generated during arc welding.
8. In arc welding, what is the magnitude of current and voltage?
Answer: in arc welding process, current values are significantly higher than voltage. Here, current varies from 60-150 A and voltage ranges from 15-25 Volts.
9. In shielded metal arc welding which type of polarity is used?
Answer: direct current reverse polarity (DCRP), here electrode is connected to the positive terminal and workpiece relates to negative terminal.
10. Which type of electrode is used in resistance spot welding?
Answer: copper electrodes (non-consumable and water cooled)
11. What are welding parameters for arc welding?
Answer: Voltage, current and feed speed
12. What is submerged arc welding?

Answer: A process in which arc is maintained under a blanket of flux

13. Arc length in arc welding should be equal to....

Answer: Rod diameter

14. In arc welding, if arc is too short, it will result in.....

Answer: Electrode sticking to the base metal and base metal not melting and bead resting on top of the work, leading to poor fusion and gas and slag holes.

15. In arc welding, if arc is too long, it will result in.....

Answer: Formation of large globules in an irregular pattern because of wandering of arc, leading to poor fusion with base plate.

16. Too low welding current in arc welding would result in.....

Answer: Excessive piling up of weld metal, poor penetration, wasted electrodes

17. Too high welding current in arc welding would result in.....

Answer: Excessive spatter, under cutting along edges, irregular deposits, wasted electrodes

18. Too fast welding speed in arc welding would result in.....

Answer: Too small bead, weak weld, and wasted electrode

19. Too slow welding speed in arc welding would result in.....

Answer: Excessive piling up of weld metal, overlapping without penetration of edges, wasted electrodes

20. In resistance welding the electrode material is made of.....

Answer: Copper

21. Which type of electrode is used in submerged arc welding.....

Answer: Bare rods

22. Flash butt welding is.....

Answer: Resistance welding